A_GMX: adaptive Gaussian-mixture clustering for skipper-CCD images

what every step does, the mathematics behind it, and how it performs against WADERs

comm_scripts clustering study

23–25 October 2026

Code: `clustering_tests/adaptive.py` (`adaptive_cluster(..., engine="gmx")`) and `clustering_tests/gmx.py` (`general_cluster`). Figures are produced by `make_agmx_figures.py` (every algorithm step, run on a real image) and `make_agmx_perf.py` (performance). All numbers are recomputed from the saved results in `~/Documents/comm_scripts_data/session_2026-10-23_clustering/clustering_2026-10-25/`.

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1 Summary

What it is. A_GMX clusters a calibrated skipper-CCD image (pixel values in electrons) without deciding pixel by pixel "in or out". It describes the image as **empty background + a sum of Gaussian charge clouds**, fits all clouds at once, shares every pixel's charge among the clouds in proportion to what each predicts there, and then groups clouds into physical deposits with two tests: a geometric one (is there a dip between two clouds?) and a statistical one (is this region several round X-rays, or one irregular deposit?). A single switch in front of it hands images whose background is *not* dark current + read noise to WADERs-like hysteresis clustering.

Why. WADERs' fixed thresholds (10 and 30 times the pedestal sigma) cut the tails of diffuse clusters (energies up to 8 % low on average, 26 % on the worst CCD), merge X-rays closer than ∼6 px, cannot see few-electron events, and on a crowded CCD every threshold is a compromise between merging and cutting.

How it does (details in section 14):

	WADERs	A_GMX
Fe-55 energy of injected clusters, width 0.3 → 1.8 px (median of 8 CCDs)	0.997 → 0.921	1.000 → 1.000
worst CCD at width 1.8 px	0.737	0.994
real Fe-55 Kα (front side): mean / spread over 8 CCDs	0.9928 / 1.32 %	0.9968 / 0.23 %
two diffuse Fe-55 clusters 5 px apart resolved	0.00	0.93
wide test, 878 image-CCDs: Fe-55 / muon / worm / 4-px pair success	0.86 / 0.59 / 0.74 / 0.11	0.93 / 0.64 / 0.79 / 0.87
Fe-55 energy resolution (rms of E/E_{true}), wide test	0.52 %	0.19 %
crowded Am-241 CCD (29 images): Np Lβ / Lγ / γ 26.3 / Ag K significance	15.5 / 16.0 / 14.8 / 12.9 σ	19.9 / 19.1 / 19.9 / 18.3 σ
crowded CCD energy scale k (1 = right)	0.984	1.001
few-electron events, 0.001 e^-/pix: 3-e^- found / fakes per Mpix	0.00 / 0	0.81 / 0
few-electron events, 0.01 e^-/pix: 5-e^- found / fakes per Mpix	0.11 / 0	0.93 / 0
run time per image-CCD (median)	0.01 s	0.8 s

Recommendation (section 16). A_GMX supersedes MHC, the hybrids and the adaptive-MHC variant as the general clusterer. WADERs stays the pipeline reference (and is what A_GMX itself uses on "unmodelled" images); plain MFD+G stays the fastest choice when only isolated X-rays matter (Fe-55 calibration).

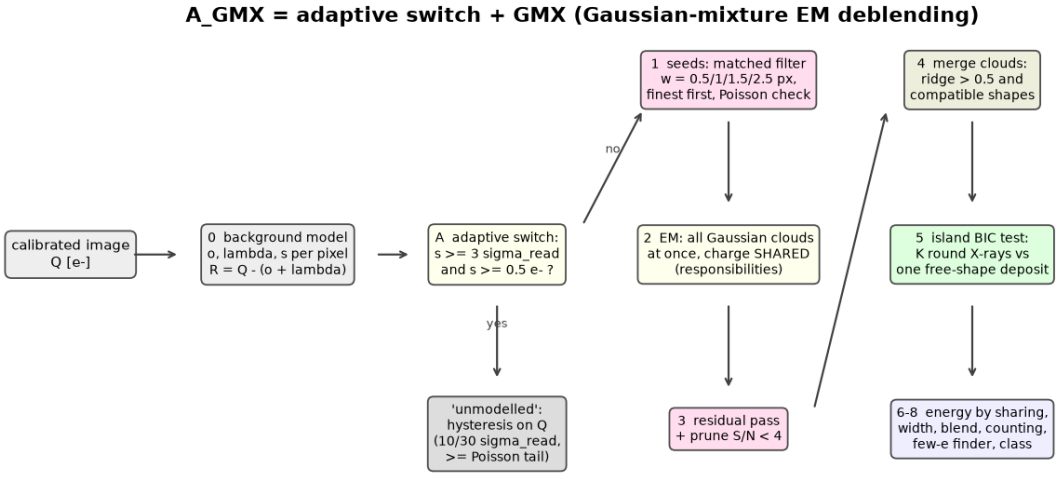


Figure 1: The steps of A_GMX (the numbers refer to the sections below).

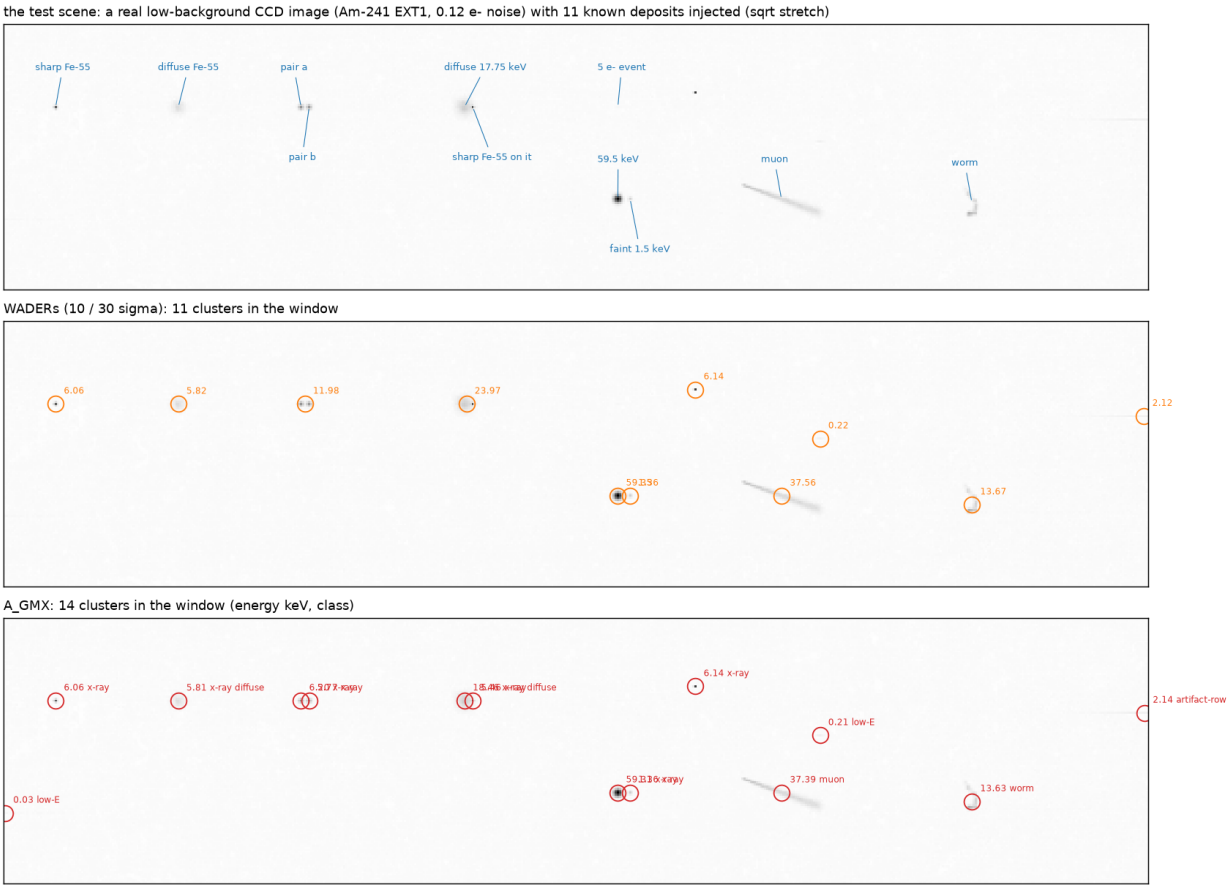


Figure 2: A real low-background CCD image (Am-241, EXT1, $0.12\,e^-$ read noise) with eleven known deposits injected (top). WADERs (middle) merges the 4-px pair into one 12-keV cluster and the sharp X-ray sitting on a diffuse one into one 24-keV cluster. A_GMX (bottom) separates both and labels the muon, the worm and the X-rays. Numbers: energies in keV.

2 The image model and notation

A calibrated image is a grid of pixel values Q_p in electrons (p = pixel). Each pixel holds

$$Q_p = o_p + n_p + \epsilon_p + D_p, \qquad n_p \sim \text{Poisson}(\lambda_p), \quad \epsilon_p \sim \mathcal{N}(0, s_p^2), \tag{1}$$

- o_p: offset (where "0 electrons" sits after calibration),
- n_p: dark charge, an integer number of electrons, Poisson with mean λ_p (e^-/pix),
- ϵ_p: read noise, Gaussian with width s_p (0.1–0.3 e^- on skipper CCDs),
- D_p: charge from particle deposits — what we want.

A deposit of charge A (electrons) whose charge cloud has covariance Σ (diffusion in the silicon: $\sim$0.3 px on the front side, up to $\sim$2 px on the back side) puts on average

$$D_p = A \int_{\text{pixel } p} \mathcal{N}(\mathbf{u}; \boldsymbol{\mu}, \Sigma)\, d^2u \tag{2}$$

electrons in pixel p. Energy = charge $\times$ 3.74 eV/e^-.

symbol	meaning
$R_p = Q_p - (o_p + \lambda_p)$	residual charge: $\approx 0\pm$ noise if empty, $\approx$ deposit charge otherwise
$V_p = \lambda_p + s_p^2$	variance of an empty pixel
$R_p^+ = \max(R_p, 0)$	the non-negative charge used as weight in the fit
$k = 1 \dots K$	clouds (components of the mixture)
$A_k,\ \boldsymbol{\mu}_k,\ \Sigma_k$	charge, centre, covariance of cloud k
$g_k(p)$	expected charge of cloud k in pixel p
$M_p = \sum_k g_k(p)$	model of the deposit charge in pixel p

3 Step 0 — the background model

Before looking for deposits, A_GMX describes the empty image (`agent_methods.background_map`). For every row, the pixels away from clusters are fitted with the **Poisson $\otimes$ Gaussian comb**

$$P(Q) = \sum_{n=0}^{\infty} \frac{\lambda^n e^{-\lambda}}{n!}\, \mathcal{N}\big(Q;\, o + n,\, s^2\big)\,, \tag{3}$$

i.e. integer electrons smeared by read noise (figure 3, bottom). The row values of o, λ, s are smoothed along the rows and in column blocks; pixels belonging to clusters are masked by a Poisson-tail test and the fit is iterated. This gives per-pixel maps o_p, λ_p, s_p (figure 3, top), hence

$$\text{mean}_p = o_p + \lambda_p, \qquad V_p = \lambda_p + s_p^2, \qquad R_p = Q_p - \text{mean}_p. \tag{4}$$

Hot columns. A column whose empty pixels are on average above the model by more than 5$\times$ the error of that mean (and by more than 0.05 e^-) gets that excess subtracted: $R_p \leftarrow R_p - \Delta_{\text{col}(p)}$.

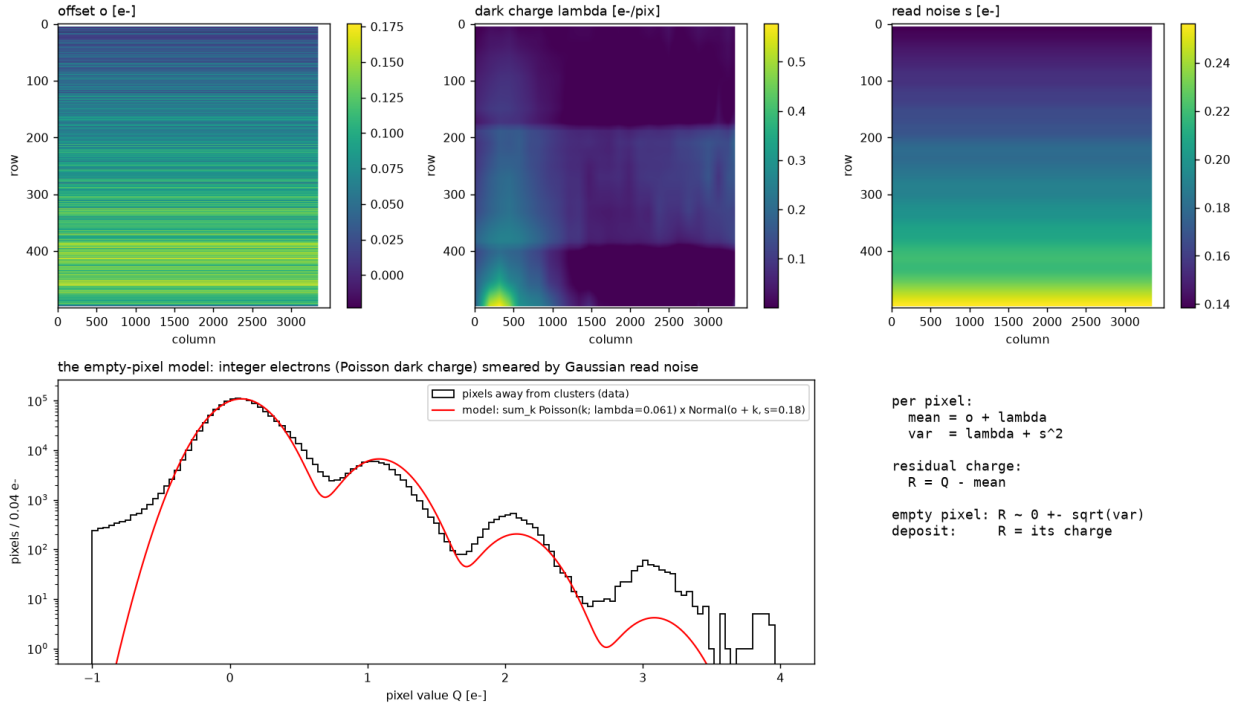


Figure 3: Top: the fitted offset, dark-charge and read-noise maps of the test image (dark charge grows along the readout). Bottom: pixels away from clusters (black) and the Poisson ⊗ Gaussian model (red): the peaks at 0, 1, 2 e^- are the integer electrons.

4 Step A — the adaptive switch

The background model can fail: on some images (here the PD06/DM12 Am-241 frames) the pixels scatter far more than dark charge + read noise allow — the fitted s is 1.5–3 e^- while the zero-electron peak is 0.37 e^- wide — and every model-based method then measures energies 5–10 % high. A_GMX therefore compares two estimates of the noise:

- **read noise from the zero peak:** $\sigma_{\text{read}} = 1.4826 \cdot \text{median}\,|Q - \text{median}(Q)|$ over the pixels with $|Q| < 0.5\,e^-$ (the median absolute deviation scaled to a Gaussian sigma; robust against pixels with charge);
- **fitted spread:** $s = \text{median}_p\, s_p$.

$$\boxed{s \geq 0.5\,e^- \quad \text{and} \quad s \geq 3\,\sigma_{\text{read}} \implies \text{“unmodelled”}} \tag{5}$$

Unmodelled → hysteresis clustering on the image itself (not on R, because the model's offset is not trusted either): pixels with $Q_p > \max(10\sigma_{\text{read}}, k{-}0.5)$ that touch (8 neighbours) form a cluster, kept if one pixel exceeds $\max(30\sigma_{\text{read}}, 9.5)\,e^-$; k is the smallest integer with $P(\text{Poisson}(\lambda) \geq k) \leq 0.10$; energy $= \sum Q_p$ over the cluster. This is WADERs with the thresholds bounded below by the dark-charge statistics.

Otherwise → GMX on the whole image (steps 1–8). In the wide test only the 27 PD06/DM12 image-CCDs of 878 were “unmodelled” (figure 4).

A_GMX also reports for every cluster the **event occupancy** of its 31 × 31-px neighbourhood: the fraction of pixels above $\max(1.5,\ 4\sigma_{\text{read}},\ k_{10^{-3}}(\lambda_p) - 0.5)$, i.e. holding *event* charge rather than dark charge ($\geq 0.3 \rightarrow$ labelled “crowded”). It is information only; GMX needs no crowded/sparse split (section 6.5 explains why).

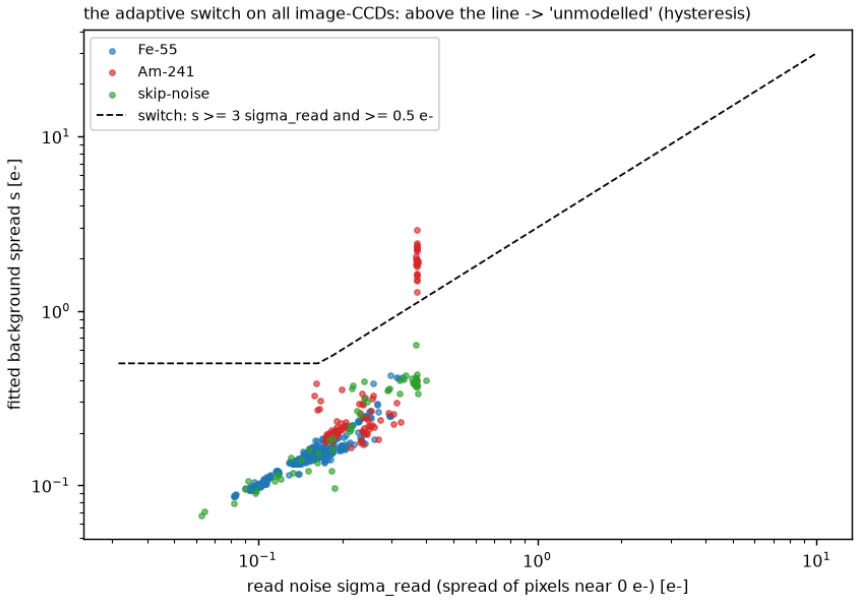


Figure 4: Fitted background spread s against the read noise σ_{read} for all 878 image-CCDs. Above the dashed line the background is not dark current + read noise ("unmodelled"): only the PD06/DM12 Am-241 images.

5 Step 1 — where are the deposits?

5.1 The matched filter

To decide whether there is a blob of width w centred at pixel c, the best *linear* statistic (largest signal-to-noise for a known shape in independent Gaussian noise) is the **matched filter**: a weighted sum of the residuals with the expected shape as weights. With $h_w(p-c)$ the pixel-integrated unit Gaussian of width w,

$$S_w(c) = \sum_p h_w(p-c)\, R_p, \qquad \mathrm{Var}[S_w(c)] = \sum_p h_w(p-c)^2 V_p, \qquad \boxed{T_w(c) = \frac{\sum_p h_w(p-c)\, R_p}{\sqrt{\sum_p h_w(p-c)^2\, V_p}}} \tag{6}$$

T_w is the significance (in Gaussian sigmas) of a blob of width w at c. Summing many pixels with the right weights makes a deposit far more significant than any single pixel: a 5.9-keV X-ray spread over 20 pixels is invisible to a per-pixel 10σ threshold in its tails but has $T \gg 100$. Both sums are separable Gaussian convolutions (`gmx._sep_conv`), computed for four widths $w \in \{0.5, 1, 1.5, 2.5\}$ px (sharp front-side X-rays to diffuse back-side ones and blobs).

5.2 Scale-space seeds

Seeds are local maxima (5×5 neighbourhood) with $T_w > 5$. The scales are visited **finest first**; a maximum at a coarser scale is kept only if no finer-scale seed lies within $\max(2, 1.5w)$ px; within a scale, seeds closer than that are suppressed (greedy, strongest first). Why: two X-rays 4–5 px apart give two maxima at $w = 0.5$ but one merged maximum in the middle at $w = 2.5$; finest-first keeps the two (scale-space detection, Lindeberg 1994). A lone diffuse X-ray has no fine-scale maximum (its peak is too flat), so its coarse-scale seed survives.

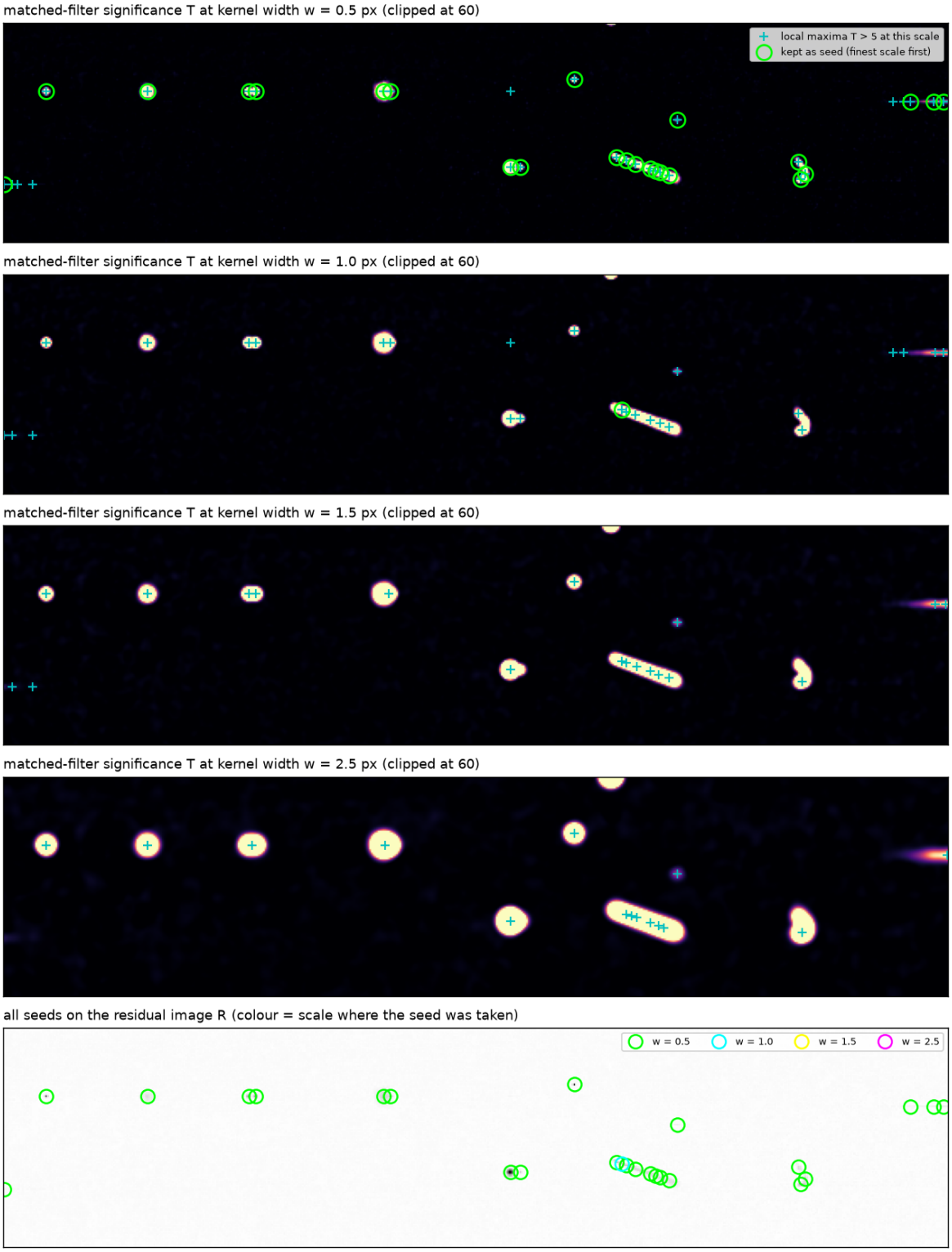


Figure 5: T_w at the four widths (clipped at 60) with the local maxima (+) and the seeds kept (circles). Bottom: all seeds on R, coloured by the scale they were taken at. The 4-px pair and the sharp X-ray on the diffuse one get two seeds each; the muon and the worm get several.

5.3 The Poisson check

T_w treats the noise as Gaussian, but dark charge is integer and Poisson: at low λ a lone box of 3 electrons is "6σ" in Gaussian terms but has probability $\sim 10^{-4}$ (figure 6). Every seed must therefore also pass an **exact counting test** on the 3×3 and the 5×5 box around it (the idea of

Li & Ma 1983 in γ-ray astronomy): with $S = \sum_{\text{box}}(Q_p - o_p)$, $\Lambda = \sum_{\text{box}} \lambda_p$, $\sigma_{\text{box}} = \sqrt{\sum_{\text{box}} s_p^2}$,

$$P(\text{Poisson}(\Lambda) \;\geq\; \lfloor S - 2\sigma_{\text{box}} + 0.5 \rfloor) < \Phi^c(5) = 2.9 \times 10^{-7} \tag{7}$$

(for either box). The $-2\sigma_{\text{box}}$ term protects against read noise pushing a box up by a fraction of an electron.

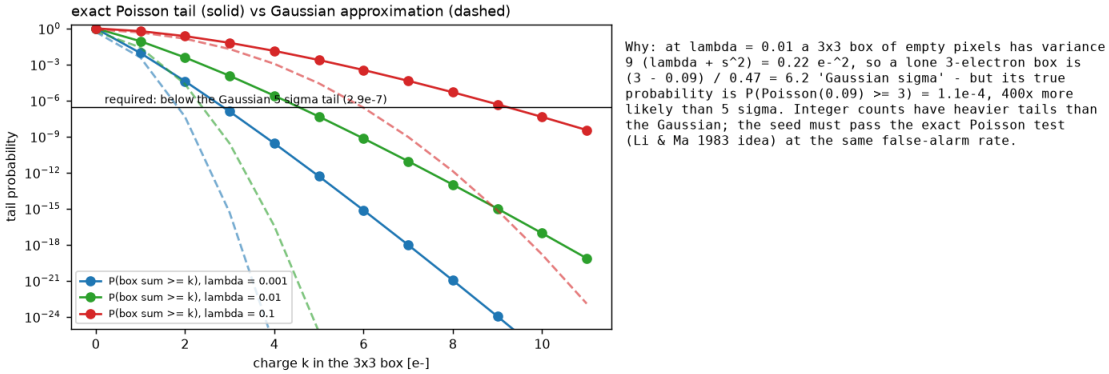


Figure 6: Tail probability of the 3×3-box charge from dark current alone: exact Poisson (solid) versus the Gaussian approximation (dashed). At low dark charge the Gaussian approximation is far too optimistic; the seed has to be below the horizontal line in the exact calculation.

Each surviving seed starts a round cloud: $\boldsymbol{\mu}$ = the seed pixel, $\Sigma = w^2\mathbf{1}$ (w = scale of the seed) and $A_0 = R_{\text{peak}} \cdot 2\pi(w^2 + \frac{1}{12})$ (the charge of a Gaussian with that peak height).

6 Step 2 — fitting all deposits at once

6.1 The model

$$M_p = \sum_{k=1}^{K} g_k(p), \qquad g_k(p) = A_k\, \mathcal{N}\big(\mathbf{x}_p;\ \boldsymbol{\mu}_k,\ \Sigma_k + \tfrac{1}{12}\mathbf{1}\big)\,. \tag{8}$$

Each cloud has its own charge A_k, centre $\boldsymbol{\mu}_k$ and **full** covariance $\Sigma_k = \begin{pmatrix} \sigma_{yy} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{xx} \end{pmatrix}$, so it can be round (an X-ray) or elliptical and tilted (a piece of a track). The $+\frac{1}{12}$ is the variance of a uniform distribution on a pixel: evaluating a Gaussian at the pixel centre with its variance increased by $\frac{1}{12}$ approximates the pixel integral (good for $\sigma \gtrsim 0.3$ px; for very sharp clusters the model's peak is too high — figure 9 shows it — but the energies do not use the model's peak, see section 10). Each cloud is evaluated only within its 4σ ellipse (at least 2 px, at most 16 px), as sparse arrays.

6.2 What is maximised

Treat the deposit charge as **counts** with mean $M_p + \varepsilon_p$, where $\varepsilon_p = \sqrt{V_p}$ is a fixed "unexplained" level per pixel (a noise component of the mixture). The Poisson log-likelihood is

$$\ln L = \sum_p \Big[R_p^+ \ln\big(M_p + \varepsilon_p\big) - \big(M_p + \varepsilon_p\big) \Big]. \tag{9}$$

Maximising it over $\{A_k, \boldsymbol{\mu}_k, \Sigma_k\}$ is a missing-data problem: if we knew which cloud each electron came from, each cloud could be fitted alone. **Expectation–maximisation** (Dempster, Laird & Rubin 1977; for Poisson images Shepp & Vardi 1982, Richardson 1972, Lucy 1974) alternates two

steps. For the amplitudes with the shapes held fixed, the update $A_k \leftarrow \sum_p r_{pk} R_p^+$ is the classic ML-EM (Richardson–Lucy) step and never decreases $\ln L$ (it assumes each cloud's profile sums to 1 over its footprint, true to $< 0.1\,\%$ for a 4σ footprint); the centre and covariance updates are the weighted-moment M-step of a Gaussian mixture. With the safeguards below (clipping, step limit) monotonic convergence is not guaranteed in general. Measured on the test image: after 30 iterations individual clouds inside a deposit still trade charge (30 more iterations change a cloud's A by a median 1.5 %), but the **cluster energies**, which share the measured charge, are stable: 60 instead of 30 iterations changes X-ray cluster energies by $< 0.004\,\%$ (99th percentile) and the median of all clusters above 0.5 keV by 10^{-6}; 2.6 % of clusters (extended or overlapping ones) change by more than 1 %.

6.3 E-step: who owns each pixel?

$$\boxed{r_{pk} = \frac{g_k(p)}{M_p + \varepsilon_p}} \qquad \left(\sum_k r_{pk} + \tfrac{\varepsilon_p}{M_p+\varepsilon_p} = 1\right) \tag{10}$$

r_{pk} is the expected fraction of pixel p's charge that came from cloud k (its **responsibility**). The noise share $\varepsilon_p/(M_p + \varepsilon_p)$ is ≈ 1 far from every cloud, so noise pixels are not handed to any cloud (figure 9, right, dotted).

6.4 M-step: update each cloud from its share

With weights $w_{pk} = r_{pk} R_p^+$ (the charge of pixel p attributed to cloud k):

$$A_k = \sum_p w_{pk}, \qquad \boldsymbol{\mu}_k = \frac{\sum_p w_{pk}\,\mathbf{x}_p}{A_k}, \qquad \Sigma_k = \frac{\sum_p w_{pk}(\mathbf{x}_p - \boldsymbol{\mu}_k)(\mathbf{x}_p - \boldsymbol{\mu}_k)^\top}{A_k} - \tfrac{1}{12}\mathbf{1}. \tag{11}$$

The charge of a cloud is the sum of its shares; its centre and covariance are the weighted moments of its shares (the $-\frac{1}{12}$ removes the pixel size again). Safeguards: variances clipped to $[0.02, 25]\,\mathrm{px}^2$, correlation $|\sigma_{xy}| \le 0.95\sqrt{\sigma_{yy}\sigma_{xx}}$, a centre moves at most 1.5 px per iteration, clouds with $A_k < 10^{-3}\,e^-$ are removed. 30 iterations.

6.5 Why this is the heart of A_GMX

No threshold decides which pixel belongs to which cluster. Where two deposits overlap, each pixel's charge is split in proportion to what each cloud predicts there. Three consequences:

1. **No percolation.** In a crowded field, threshold-and-connect methods chain neighbours into one blob (or need such high thresholds that tails are cut). Sharing has no such failure mode: this is why A_GMX beats WADERs on the crowded Am-241 CCD.

2. **No charge stealing.** A sharp X-ray on the shoulder of a diffuse one keeps only its own charge (figure 9).

3. **Tails are kept.** The faint tail of a diffuse cluster is explained by its Gaussian, not cut by a threshold: energies are unbiased at every width (figure 17).

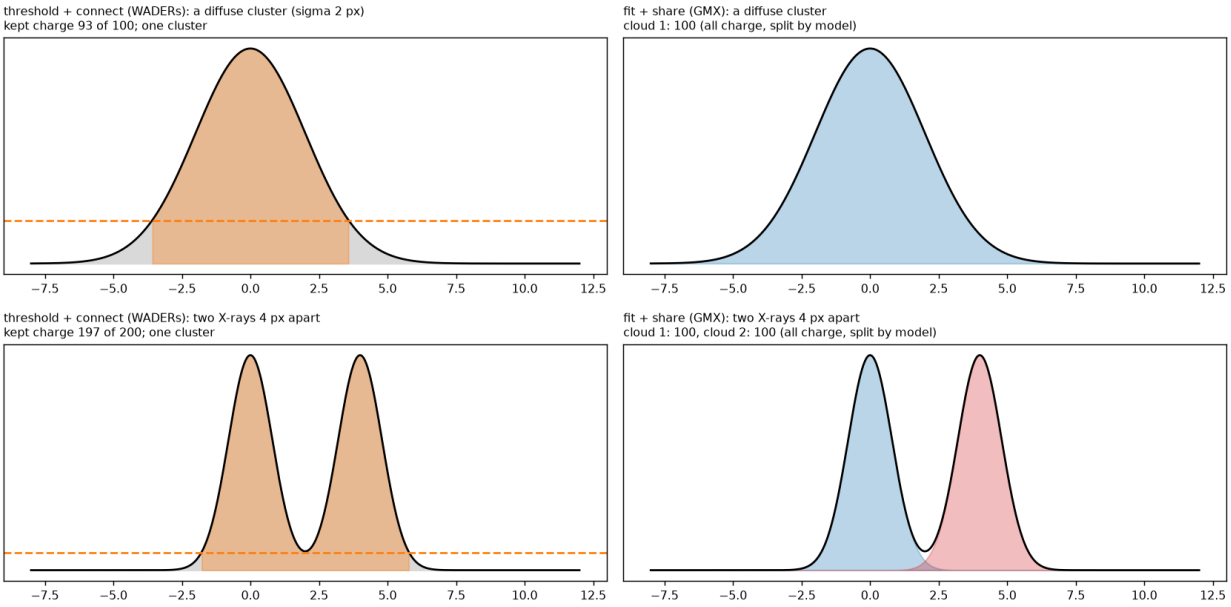


Figure 7: The difference in one dimension. Left, threshold + connect: the tails of a diffuse cluster below the threshold are lost (93 of 100 e^- kept here) and two X-rays 4 px apart are connected into one cluster. Right, fit + share: every electron is kept and the overlap region is split between the two clouds in proportion to their predicted charge.

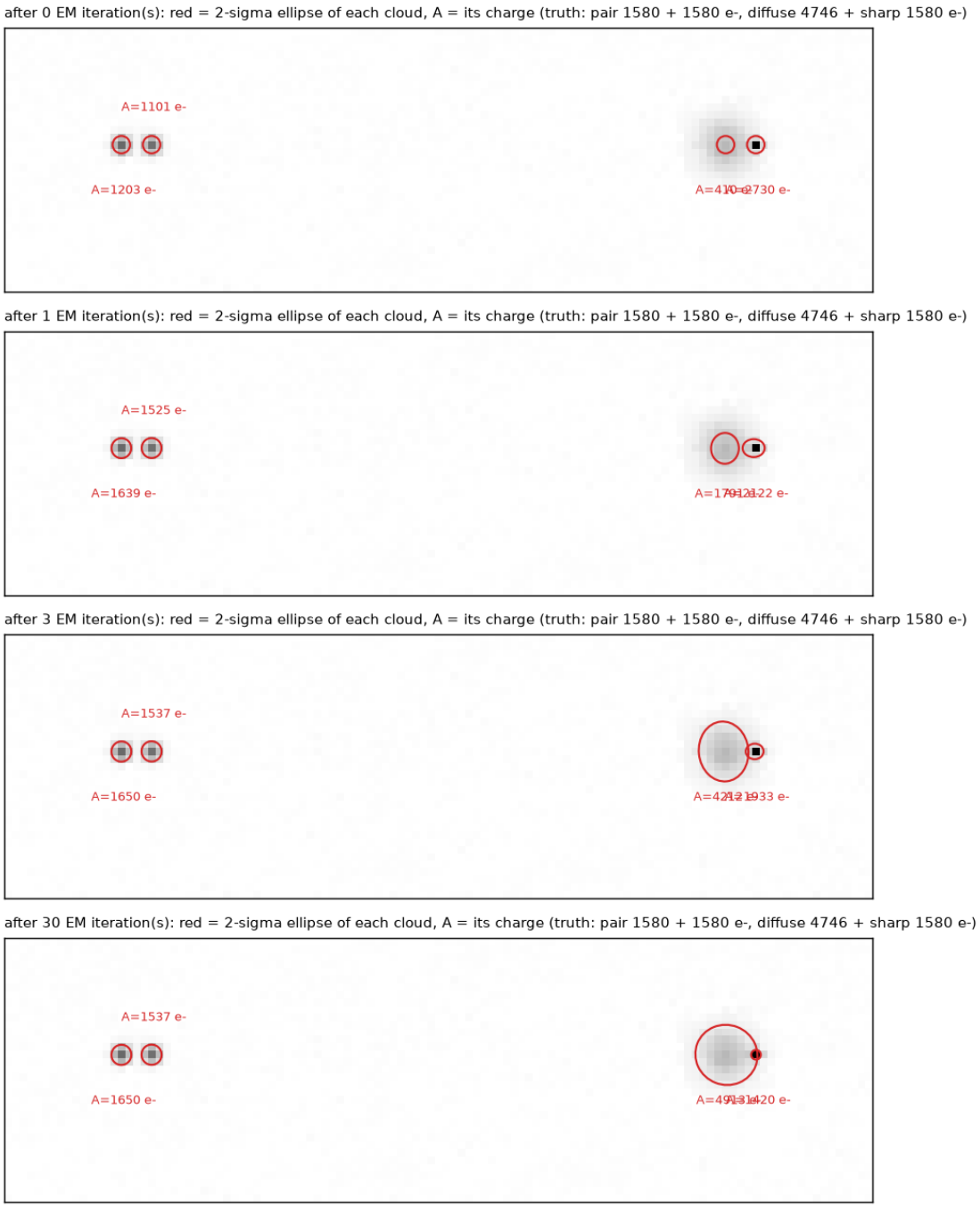


Figure 8: EM on the 4-px pair (left) and the sharp X-ray on a diffuse one (right): 2σ ellipses and charges A after 0, 1, 3 and 30 iterations. The diffuse cloud grows to its true width, the sharp one shrinks onto its pixel, and the charges converge to the truth ($1580 + 1580\,e^-$; $4746 + 1580\,e^-$).

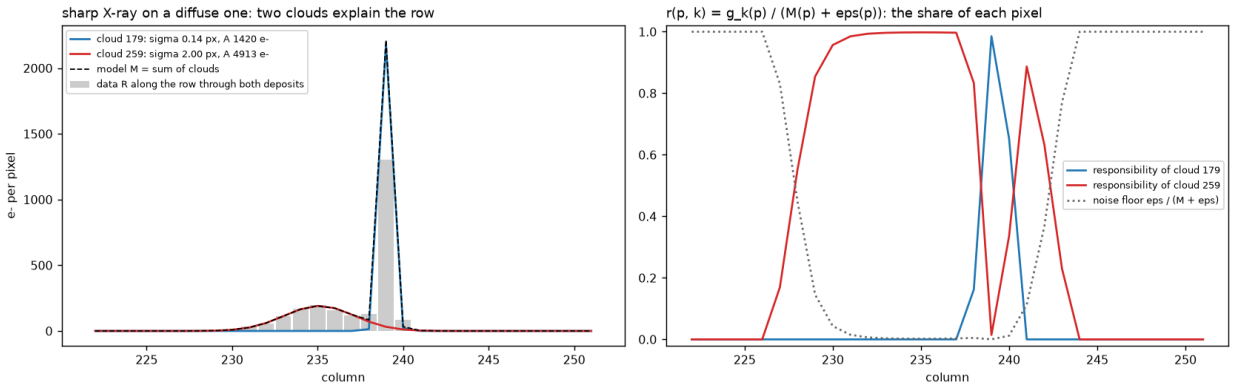


Figure 9: Left: the row through the sharp X-ray on the diffuse one — data (grey), the two clouds and their sum. Right: responsibilities; each pixel's charge is split between the two clouds, and far from both the noise share is ∼1.

7 Step 3 — residual pass and pruning

Residual pass. A faint deposit right next to a bright one can be hidden at the seed stage. After EM, the matched filter is run on the residual $R - M$ with the model's own Poisson variance added, $V'_p = V_p + M_p$, at the two finest scales; local maxima (3×3) with $T > 5$ more than 1 px from any cloud become new clouds, and EM runs 15 more iterations on all clouds (figure 10).

Pruning. The significance of cloud k is its charge divided by the uncertainty of that charge. For a linear fit of A_k with known profile $\pi_k(p) = g_k(p)/A_k$ and per-pixel variance v_p (read noise + dark charge + the other clouds' model, i.e. the variance if this cloud were absent), the Fisher information is $\sum_p \pi_k(p)^2/v_p$, so

$$\mathrm{S/N}_k = A_k \sqrt{\sum_p \frac{\pi_k(p)^2}{V_p + \left(M_p - g_k(p)\right)}}. \tag{12}$$

Clouds with $\mathrm{S/N} < 4$ are dropped; EM runs 5 more iterations.

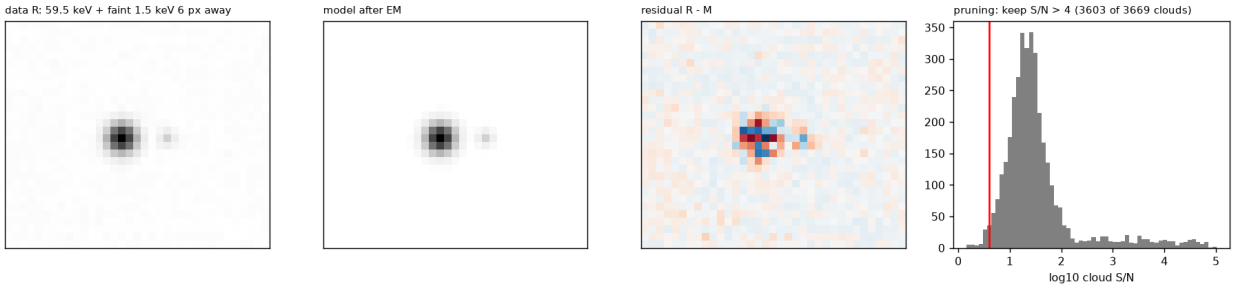


Figure 10: A 59.5-keV X-ray with a faint 1.5-keV X-ray 6 px away: data, model after the first EM, residual (the faint one stands out; circle = residual seed), and the S/N of all clouds of the image with the pruning cut.

8 Step 4 — which clouds are one deposit?

One deposit can need several clouds (a track needs many; a non-Gaussian X-ray may need two). Clouds are grouped by testing every pair closer than $3.5(\bar{\sigma}_i + \bar{\sigma}_j) + 1$ px ($\bar{\sigma}$ = geometric-mean width).

Ridge (dip) test. Walk the straight line between the two centres, $\mathbf{x}(t) = \boldsymbol{\mu}_i + t(\boldsymbol{\mu}_j - \boldsymbol{\mu}_i)$, $t \in [0, 1]$, and evaluate the two-cloud density $f(t) = g_i(\mathbf{x}(t)) + g_j(\mathbf{x}(t))$:

$$\rho_{ij} = \frac{\min_{t\in[0,1]} f(t)}{\min\big(f(0), f(1)\big)}. \tag{13}$$

$\rho \approx 1$: no valley, one connected object. $\rho \ll 1$: two separate peaks. Required: $\rho > 0.5$ (the modality criterion for merging mixture components; Ray & Lindsay 2005, Hennig 2010).

Shape test. Even without a dip, two clouds are merged only if **one** of these holds:

- similar widths: $\max(\bar{\sigma}_i, \bar{\sigma}_j)/\min(\bar{\sigma}_i, \bar{\sigma}_j) < 1.6$;
- a track: one cloud is elongated (axis ratio > 1.4) **and** its long axis points along the line joining them ($|\cos| > 0.8$);
- a shoulder: the narrower cloud is not a local maximum of f (it only shapes the flank of the broad one).

The case this protects: a **sharp X-ray on top of a diffuse one** is a maximum of its own with a very different width — kept apart. Pairs that pass are linked; the connected components of the links are the clusters.

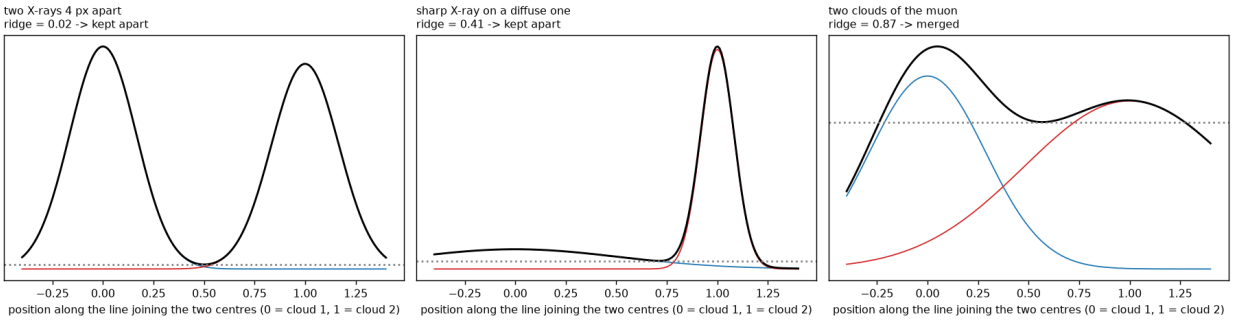


Figure 11: Two-cloud density along the line between centres. Left: two X-rays 4 px apart, deep valley ($\rho = 0.02$), kept apart. Middle: sharp X-ray on a diffuse one — no deep valley ($\rho = 0.41$) but the sharp one is a maximum of its own with a very different width, kept apart. Right: two clouds of the muon, $\rho = 0.87$, merged.

9 Step 5 — tracks versus several X-rays

Lumpy electron tracks ("worms") have real dips along them (their $\mathrm{d}E/\mathrm{d}x$ fluctuates), so the ridge test alone can split them. The **island test** asks the data directly.

Islands are connected regions (8 neighbours) of pixels above the Poisson-tail threshold of the dark charge at $\alpha = 0.01$. An island of at most 4000 px that holds $K \geq 2$ clusters (from step 4) is fitted with two hypotheses on its pixels (+2 rings):

- H_{xr}: K **separate X-rays** — one *round* Gaussian per cluster (started from the cluster's clouds, refitted by EM with $\Sigma = \sigma^2\mathbf{1}$): 4 parameters each (A, y, x, σ);
- H_{free}: **one irregular deposit** — the free-shape clouds already fitted: 6 parameters each $(A, y, x, \sigma_{yy}, \sigma_{xx}, \sigma_{xy})$, n_c clouds.

Each is scored with a weighted χ^2 and the **Bayesian information criterion** (Schwarz 1978), which charges $\ln n$ per parameter (n = pixels):

$$\chi^2_H = \sum_p \frac{\left(R_p - M_p^H\right)^2}{V_p + R_p^+ + (0.05\,R_p)^2}, \qquad \Delta\mathrm{BIC} = \left(\chi^2_{\mathrm{xr}} + 4K\ln n\right) - \left(\chi^2_{\mathrm{free}} + 6n_c \ln n\right). \tag{14}$$

The $(0.05R)^2$ term tolerates the fact that real X-rays are not perfect Gaussians (5 % model error). If $\Delta\mathrm{BIC} > 30$ — the free-shape description is so much better that it is worth its extra parameters — **the whole island becomes one deposit**. In crowded areas islands percolate beyond 4000 px and the test switches itself off (there, sharing does the work).

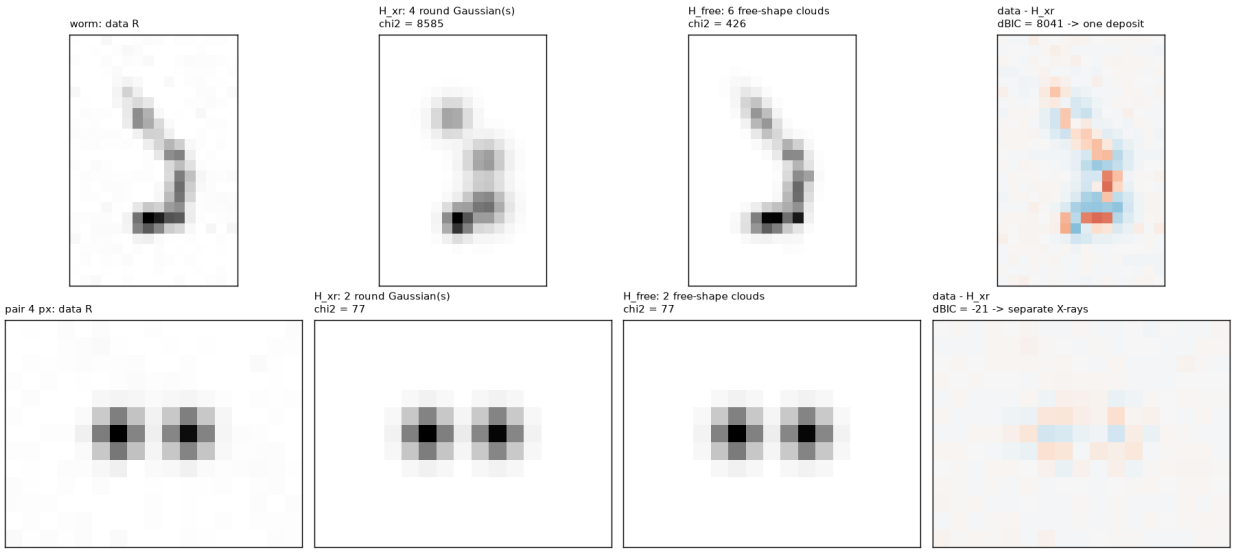


Figure 12: The island test on two islands of the test image. Top, the worm: 4 round Gaussians (one per cluster left by the ridge test) cannot follow the bent track (χ^2 8585, structured residual); 6 free-shape clouds can (χ^2 426); $\Delta\mathrm{BIC} = 8041 \rightarrow$ one deposit. Bottom, the 4-px pair: both descriptions fit equally well (χ^2 77), the round one has fewer parameters, $\Delta\mathrm{BIC} = -21 \rightarrow$ two separate X-rays.

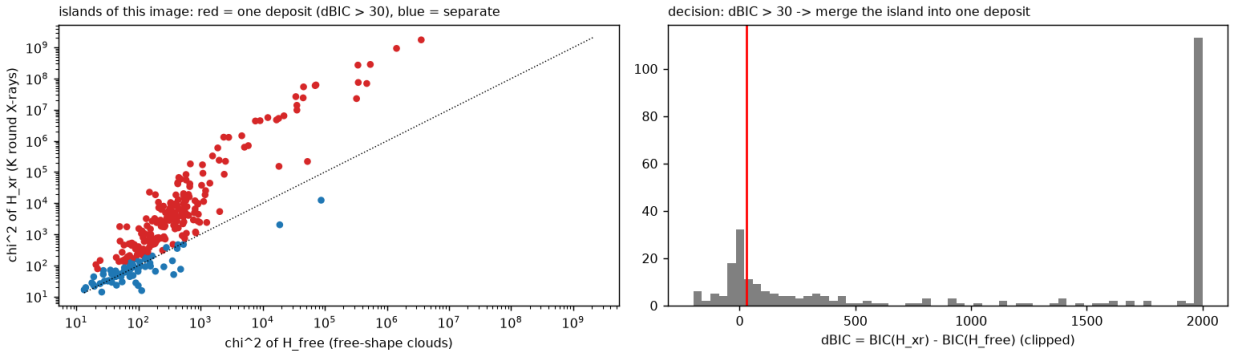


Figure 13: All islands of the test image: χ^2 of the two hypotheses (left) and $\Delta\mathrm{BIC}$ (right). Red: one deposit (tracks, worms, irregular blobs); blue: separate X-rays.

10 Step 6 — energy, shape, purity

Energy by sharing. The cluster responsibility of pixel p is $\tilde{r}_{pc} = \sum_{k \in c} g_k(p)/M_p$ (sum over the clouds of cluster c), and

$$\boxed{E_c = \sum_p \tilde{r}_{pc}\, R_p} \qquad [e^-;\ \times 3.74\ \text{eV/e}^-\ \text{for keV}] \tag{15}$$

The **measured** charge R_p is shared (not the model), so the energy does not depend on the Gaussian being exact; shared pixels are split, never counted twice.

- **Satellites.** A cluster with $E < 1\,\%$ of a neighbour's energy *and* $E < 1\,\text{keV}$ whose centre lies in that neighbour's domain or within 6σ (Mahalanobis) of one of its clouds is a tail fragment and is merged into it. (Until 25 October there was no 1-keV limit, and a real X-ray next to a very large deposit could be swallowed by it; see section 15.)
- **Growth.** Pixels above the Poisson-tail threshold ($\alpha = 0.01$) within 3 px of a covered pixel and connected to it, but covered by no cloud (CTI trails along the readout), are added to the nearest cluster.

Width. One-cloud clusters: $\sigma_{xy} = \sqrt{(\sigma_{yy} + \sigma_{xx})/2}$ of the fitted cloud; several clouds: second moment of the shared charge, $\sigma_{xy}^2 = \frac{1}{2}(\text{var}_y + \text{var}_x) - \frac{1}{12}$.

Significance. $E_c \big/ \sqrt{\sum_p \tilde{r}_{pc}^2 V_p}$.

Purity: blend. How much of *other* clusters' predicted charge lies under this cluster's profile, relative to its own:

$$\boxed{\text{blend}_c = \frac{\sum_p M_p^{(c)} M_p^{(\neg c)}}{\sum_p \left(M_p^{(c)}\right)^2}}, \qquad M_p^{(c)} = \tilde{r}_{pc} M_p, \quad M_p^{(\neg c)} = M_p - M_p^{(c)}. \tag{16}$$

0 = isolated; 0.1 = 10 % of its profile is shared with neighbours. Cutting blend < 0.02 keeps clusters whose energy is not affected by sharing — on the crowded CCD this is what turns Np Lβ 15.5σ (WADERs) into 19.9σ (figure 20). The same idea is SExtractor's deblending flag, but here it is a number. **Edge** flags clusters touching the 2-px border of the active area.

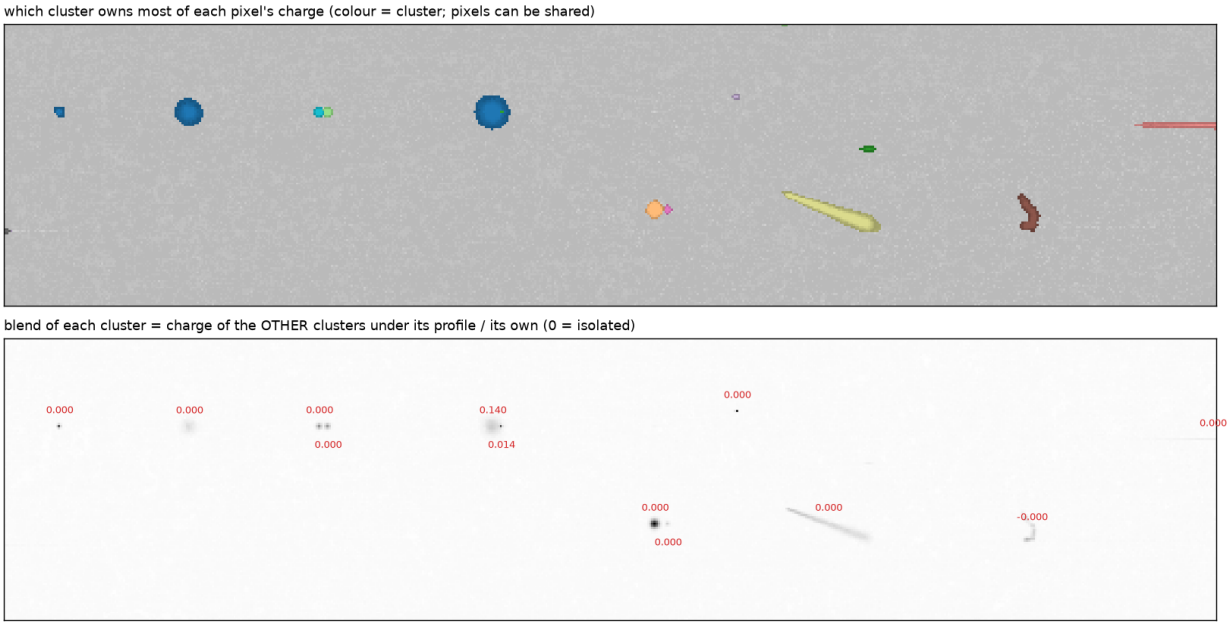


Figure 14: Top: which cluster owns most of each pixel's charge. Bottom: blend of each cluster (all ≈ 0 here except the sharp X-ray on the diffuse one, whose profile shares charge with its neighbour).

11 Step 7 — few-electron events

On photon-counting CCDs (median $s < 0.5\,e^-$) two extra rules apply.

Counting energies. For clusters below $100\,e^-$, the analog sum adds the read noise of every pixel ($\sqrt{N}\,s$, $0.5\,e^-$ for 13 pixels at $0.15\,e^-$), so the charge is often off by one. Instead each pixel the cluster mostly owns ($\tilde{r} > 0.5$) is **rounded** to whole electrons and the expected dark charge is subtracted:

$$E_c^{\text{count}} = \sum_{p:\ \tilde{r}_{pc}>0.5} \Big(\text{round}(Q_p - o_p) - \lambda_p\Big). \tag{17}$$

Counting finder. Groups of at most 9 connected pixels with $\text{round}(Q_p - o_p) \geq 1$ not covered by any cluster are added as "low-E" events when their charge is unlikely from dark current alone:

$$P\Big(\text{Poisson}\big(\textstyle\sum_{\text{group+1 ring}} \lambda_p\big) \geq S\Big) < 10^{-5}, \qquad S = \textstyle\sum_{\text{group}} \text{round}(Q_p - o_p). \tag{18}$$

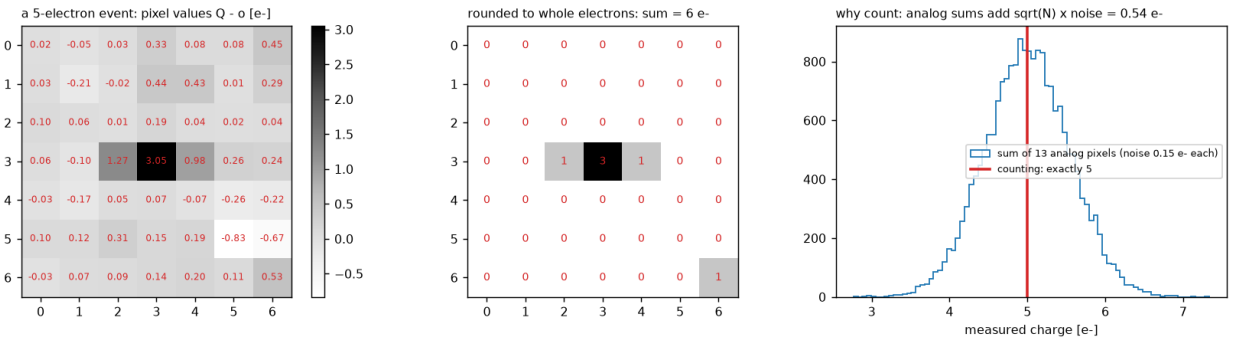


Figure 15: A 5-electron event: analog pixel values, the same rounded to whole electrons, and why counting is exact while an analog sum over 13 pixels spreads by $\pm 0.5\,e^-$.

12 Step 8 — classes

Every cluster gets a label from its shape (the rules of MHC): **low-E** ($E < 0.5$ keV); **x-ray / x-ray diffuse** (one round cloud or extent < 6 px; diffuse if $\sigma_{xy} \geq 1$ px); **alpha** (> 1 MeV); **artifact-row / -column** ($\geq 85\,\%$ of the charge in ≤ 2 rows / columns and a run of ≥ 8 px: serial-register streaks, hot columns); **muon** (long, straight skeleton: length ≥ 10 px, scatter of the cloud centres about a line < 1 px and $< 6\,\%$ of the length); **worm** (everything else extended). Labels are only checked on injected objects (alpha 0.94–1.00, x-ray 0.8, muon/worm swap $\sim$30 %).

13 The worked example

The test scene of figure 2: truth versus both methods (energies in keV).

deposit	truth	A_GMX	WADERs
sharp Fe-55	6.06	6.06 x-ray	6.06
diffuse Fe-55 (σ 1.6 px)	5.81	5.81 x-ray diffuse	5.82
pair, 4 px apart	6.20 + 5.77	**6.20 + 5.77**	11.98 (merged)
diffuse 17.75 keV + sharp Fe-55 3 px away	18.07 + 5.82	**18.46 + 5.46** (sum conserved)	23.97 (merged)
59.5 keV + faint 1.5 keV 6 px away	59.32 + 1.37	59.31 + 1.36	59.35 + 1.36
muon (40 px)	37.40	37.39 muon	37.56
worm	13.63	13.63 worm	13.67
5-electron event	0.019	not found	not found

The sharp-on-diffuse case is split, but 0.36 keV of the sharp X-ray's charge is assigned to the diffuse one: the Gaussian model of a very sharp cluster is approximate (section 6.1) and the sharing inherits that. The 5-e^- event sits on a CCD with $\sim$0.07 e^-/pix dark charge, where 5 e^- is at the detection limit (figure 19).

14 Performance against WADERs and the other methods

Five independent tests, all with the same injected objects for every method (same random seeds), plus real lines.

14.1 The wide test: every image on the machine

878 image-CCDs from 16 data sets (Fe-55 production modules DM01–DM28 and PD06–08; Am-241 AMTEst and PD06/DM12; the skip-noise study). Per image-CCD: 12 Fe-55 clusters (σ 0.3–1.6 px), 4 muons, 4 worms, 4 pairs 4 px apart. Success = one cluster with energy within 5 % (pairs: both, within 10 %).

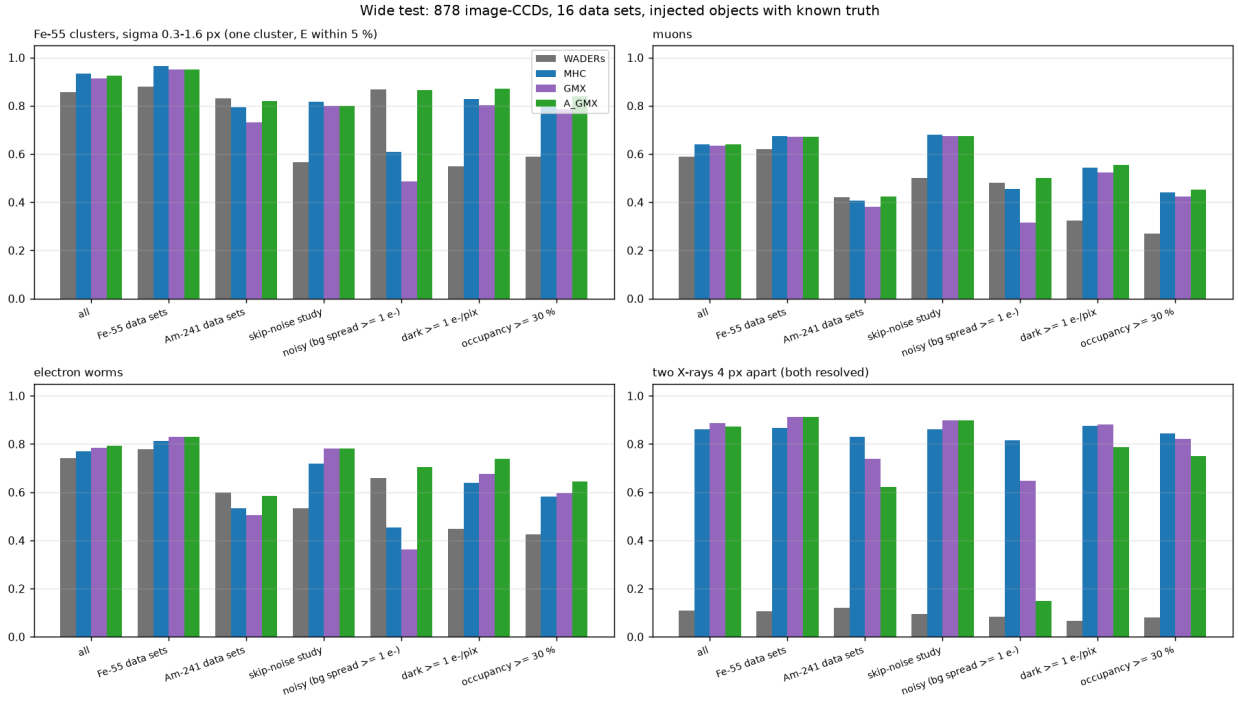


Figure 16: Success fractions by object type and image condition for WADERs, MHC, GMX and A_GMX.

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD
WADERs	0.856	0.52 %	0.59	0.74	0.11	0.01
B auto	0.905	0.59 %	0.62	0.78	0.13	0.01
MFD+G	0.933	0.21 %	0.18	0.37	0.94	0.14
hybrid v1	0.873	0.20 %	0.58	0.66	0.84	0.02
hybrid v2	0.930	0.21 %	0.63	0.68	0.85	0.05
MHC	0.933	0.21 %	0.64	0.77	0.86	1.61
adaptive (MHC engine)	0.931	0.21 %	0.64	0.77	0.83	1.26
GMX	0.913	0.19 %	0.63	0.78	0.89	0.65
A_GMX	**0.925**	**0.19 %**	**0.64**	**0.79**	**0.87**	0.81

Wide test, all 878 image-CCDs.

condition (Fe-55 success / worm / pair)	WADERs	MHC	GMX	A_GMX
Fe-55 data sets (705)	0.88 / 0.78 / 0.11	0.96 / 0.81 / 0.87	0.95 / 0.83 / 0.91	0.95 / 0.83 / 0.91
noisy, background spread $\geq 1\,e^-$ (27)	0.87 / 0.66 / 0.08	0.61 / 0.45 / 0.81	0.49 / 0.36 / 0.65	0.86 / 0.70 / 0.15
dark charge $\geq 1\,e^-$/pix (152)	0.55 / 0.45 / 0.07	0.83 / 0.64 / 0.88	0.80 / 0.67 / 0.88	0.87 / 0.74 / 0.79
event occupancy $\geq 30\,\%$ (194)	0.59 / 0.43 / 0.08	0.82 / 0.58 / 0.85	0.78 / 0.60 / 0.82	0.84 / 0.65 / 0.75

Wide test by image condition (number of image-CCDs in brackets).

14.2 Fe-55: energy versus cluster width, pairs, real Kα

DM09/DM10 (EXT1, EXT4, images 0–1; 220 injected clusters per width and CCD; pairs of σ 1.4-px clusters).

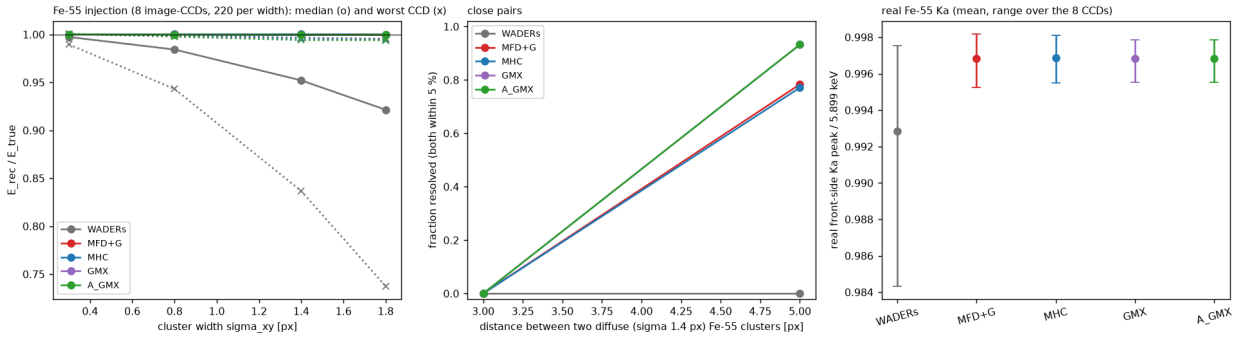


Figure 17: Left: median (circles) and worst-CCD (crosses) energy ratio against cluster width. Middle: diffuse pairs resolved. Right: real front-side Kα over the 8 CCDs.

method	σ 0.3	0.8	1.4	1.8 px	pair 5 px	real Kα/5.899	spread
WADERs	0.9974 (0.990)	0.9843 (0.943)	0.9521 (0.837)	0.9214 (0.737)	0.00	0.9928	1.32 %
MFD+G	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9999 (0.995)	0.78	0.9968	0.29 %
MHC	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9998 (0.995)	0.77	0.9969	0.26 %
A_GMX	1.0000 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	**0.93**	0.9968	**0.23 %**

Median $E_{\mathrm{rec}}/E_{\mathrm{true}}$ (worst CCD in brackets) per cluster width; pairs resolved; real Kα.

14.3 Injected tracks, alphas and artifacts in real Am-241 images

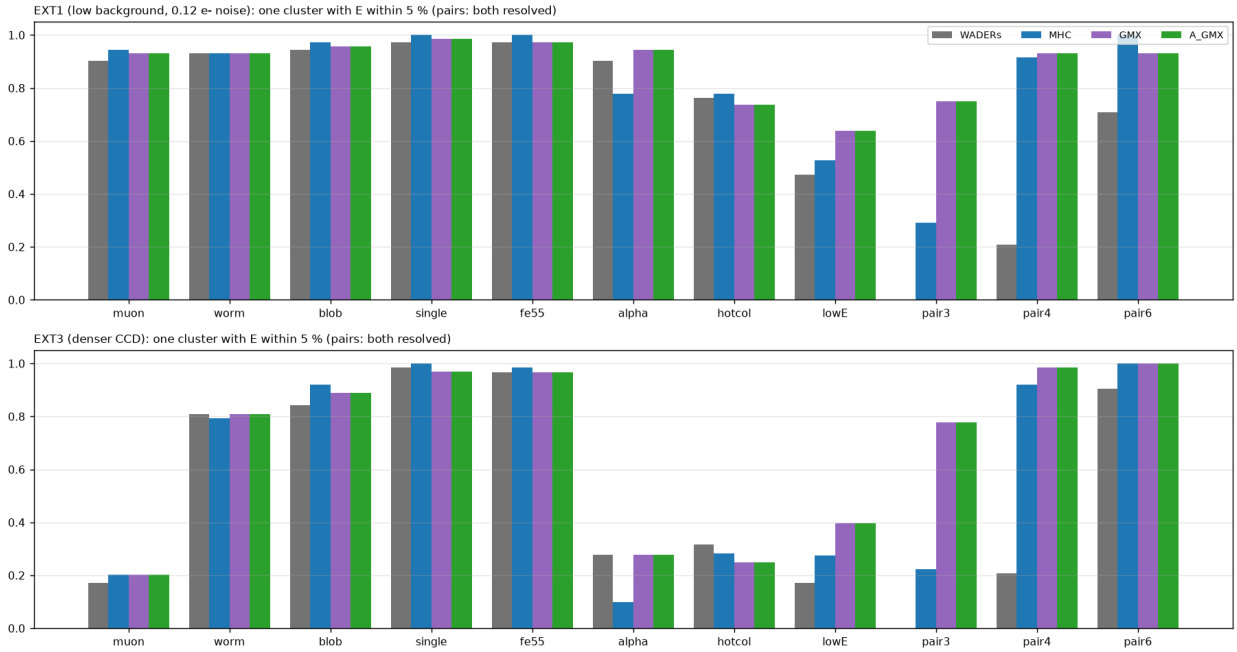


Figure 18: EXT1 (low background) and EXT3 (denser CCD): success per object type.

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
A_GMX	0.93	0.93	0.96	0.99	0.97	**0.94**	0.74	**0.64**	**0.75**	0.93	0.93

EXT1, 3 images × 3 seeds: success fraction per object type.

14.4 Few-electron events

Simulated blank images (Poisson dark charge + 0.15 e^- noise, 2 × 1.7 Mpix per dark-charge value), 800 events of 1–20 e^- in 1–5 pixels injected; fakes counted on a second blank image with nothing injected.

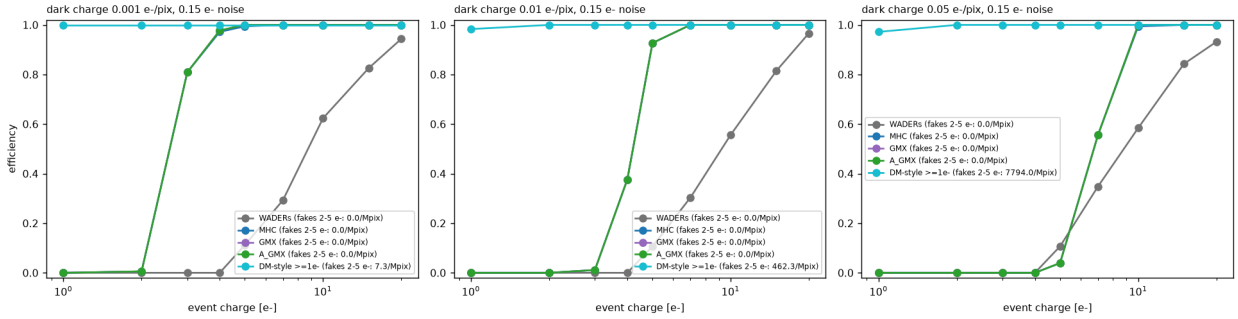


Figure 19: Efficiency against event charge at three dark-charge levels; the legend gives the fake clusters of 2–5 e^- per Mpix. A_GMX (= GMX = MHC here) finds 81 % of 3-e^- events at 0.001 e^-/pix and 93 % of 5-e^- events at 0.01 e^-/pix, with no fakes; at 0.05 e^-/pix it needs ∼7 e^- (and WADERs is slightly better at 5 e^- there: 0.11 vs 0.04 — the Poisson check is strict at that dark charge). WADERs needs $\gtrsim$ 10 e^- at low dark charge. Clustering of $\geq$ 1-e^- pixels (DM-style) finds everything, but its 2–5-e^- fakes from dark-current coincidences are 7 / 460 / 7800 per Mpix at 0.001 / 0.01 / 0.05 e^-/pix.

14.5 The crowded CCD: Am-241 10–30 keV constrained fit

29 images, EXT4 (in one corner every pixel holds > 20 e^-), Np L lines with fixed intensity ratios (Lépy et al. 2008), one energy scale k.

clustering (cut)	clusters	k	dev/ndf	Np Lα	Lβ	Lγ	γ 26.3	Ag K
WADERs ($\sigma_{xy} < 0.7$)	62 665	0.984	1.74	4.0	15.5	16.0	14.8	12.9
MFD+G ($\sigma_{xy} < 0.7$)	28 811	1.003	1.14	2.1	8.2	9.1	8.0	8.5
MHC ($\sigma_{xy} < 0.7$)	27 250	1.003	1.10	1.9	8.0	9.1	8.2	8.5
hybrid v1 ($\sigma_{xy} < 0.7$)	5 270	0.999	1.31	0.7	4.4	5.2	5.6	3.5
hybrid v2 ($\sigma_{xy} < 0.7$)	23 862	1.002	1.14	1.9	7.9	8.5	8.4	8.4
adaptive, MHC engine ($\sigma_{xy} < 0.7$)	60 158	0.986	1.51	3.0	14.4	15.4	14.0	12.0
A_GMX ($\sigma_{xy} < 0.7$)	120 280	1.001	1.74	1.3	14.6	15.6	14.6	14.1
A_GMX ($\sigma_{xy} < 0.7$, blend < 0.02)	65 876	**1.001**	1.98	**4.3**	**19.9**	**19.1**	**19.9**	**18.3**

Checks: the same result in the even and the odd images separately (combined Z 24.8 and 23.6); an equivalent isolation cut on WADERs clusters (nearest cluster >3–20 px) makes WADERs *worse* (Z 27.1 → 6.2), so the gain is the sharing, not the cut; A_GMX beats WADERs for every blend cut up to 0.3 (figure 20, right). Caveats: deviance/ndf 1.98 (vs 1.74) with structure at 25–28 keV, and σ_{xy} is a different width estimate from panaSKImg's fitted one (the 0.7 cut was not re-optimised).

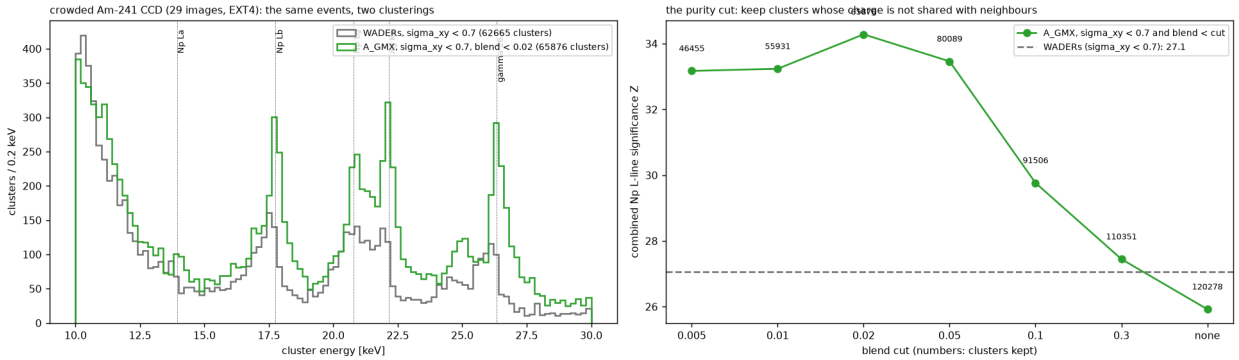


Figure 20: Left: 10–30-keV spectra of the crowded CCD, WADERs vs A_GMX with the blend cut — the same events, about twice the peak height above the continuum. Right: combined Np L significance against the blend cut.

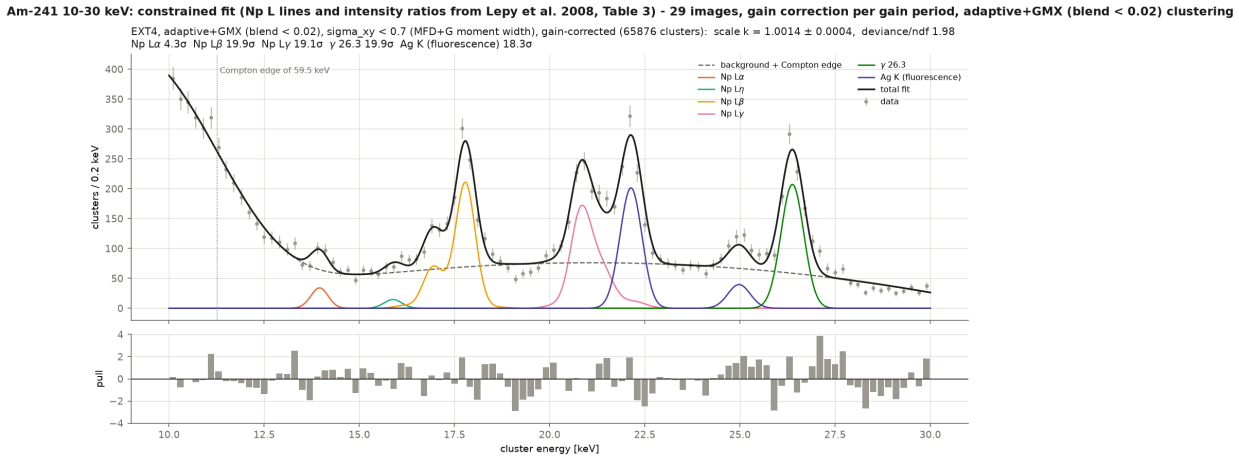


Figure 21: The constrained fit of the crowded CCD with A_GMX clusters ($\sigma_{xy} < 0.7$, blend < 0.02).

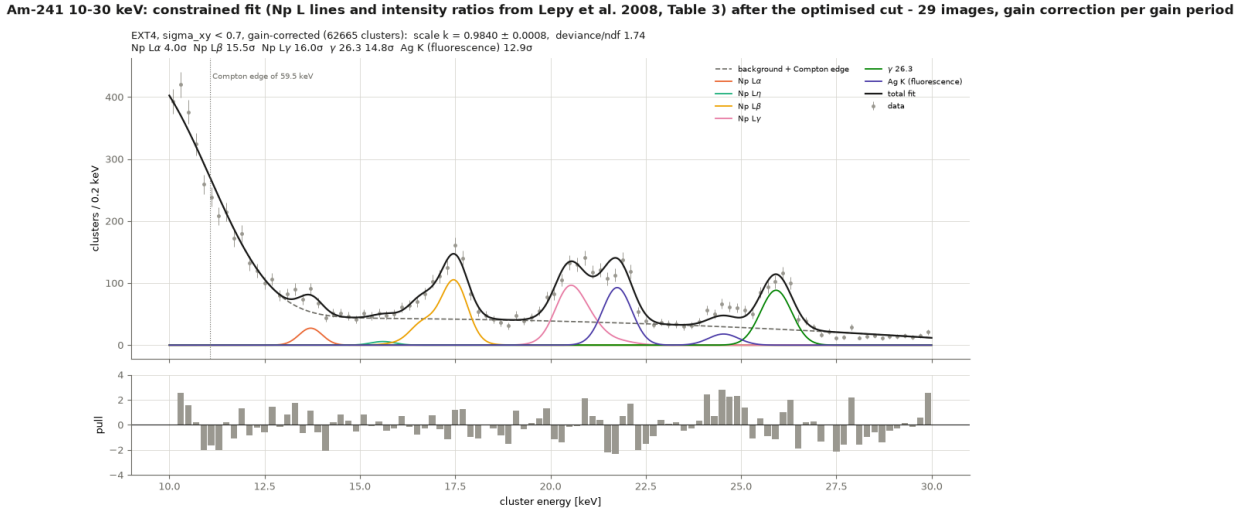


Figure 22: The same fit with the WADERs clusters ($\sigma_{xy} < 0.7$; panaSKImg's fitted width), for comparison.

14.6 Run time

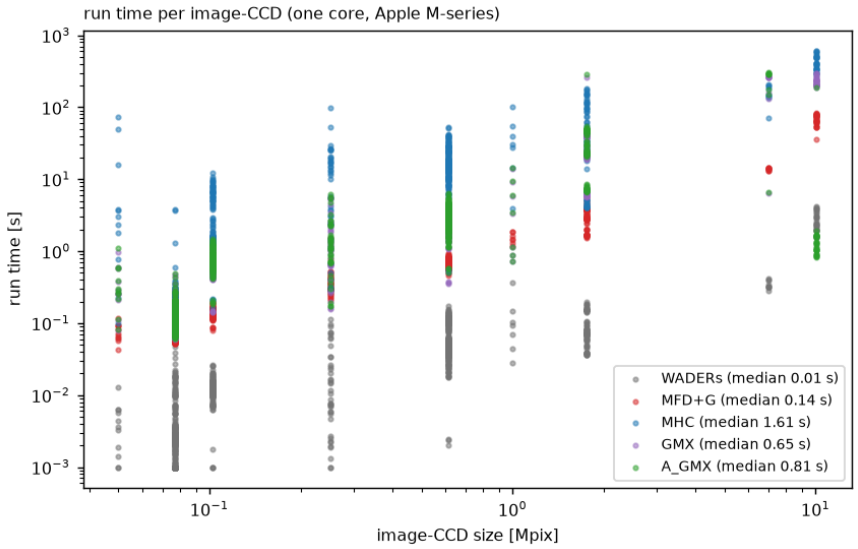


Figure 23: Run time per image-CCD against its size (one core). A_GMX: median 0.8 s; ∼1 s per Mpix on sparse images, up to ∼40 s on a crowded 1.75-Mpix CCD.

15 Contamination: the high-side tail and the quality flags (25 October)

On the crowded Am-241 CCD the 59.5 keV line of A_GMX has a high-side tail and the lines sit slightly high ($k = 1.0014$; figure 24). This section shows where it comes from, what was tried, and what changed in the code.

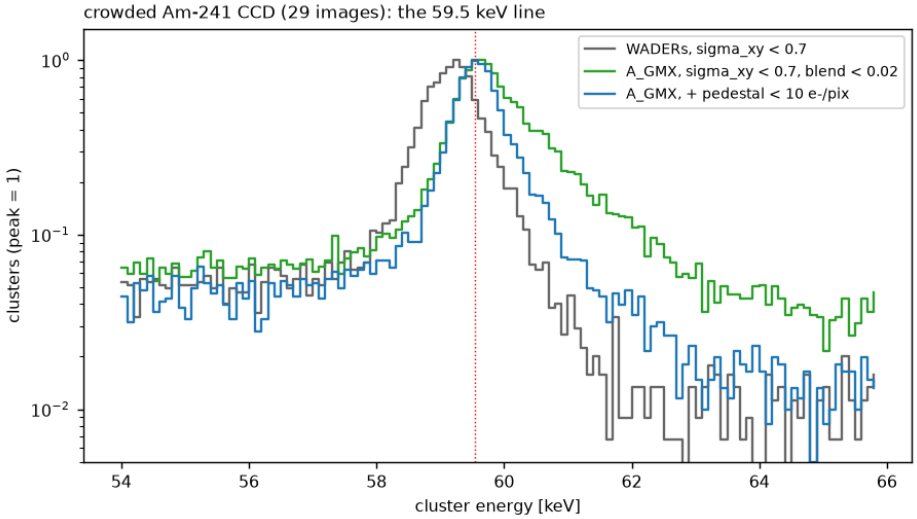


Figure 24: The 59.5 keV line of the crowded CCD (29 images): WADERs, A_GMX with the cuts of section 14, and A_GMX with the new pedestal cut. Scaled to the peak.

15.1 Is the algorithm biased? No.

Clusters of 5.9, 17.75, 26.34 and 59.54 keV were injected at *random* positions (also next to real events) into real images. Away from real events A_GMX is unbiased at every energy (median

$E_{\mathrm{rec}}/E_{\mathrm{true}} = 0.9998$, spread 0.03 % at 59.5 keV, no high tail), on the clean and the crowded CCD. Next to real events every method adds charge of the neighbour: median 1.033 (clean CCD) and 1.063 (crowded CCD) for A_GMX, similar for WADERs (figure 25).

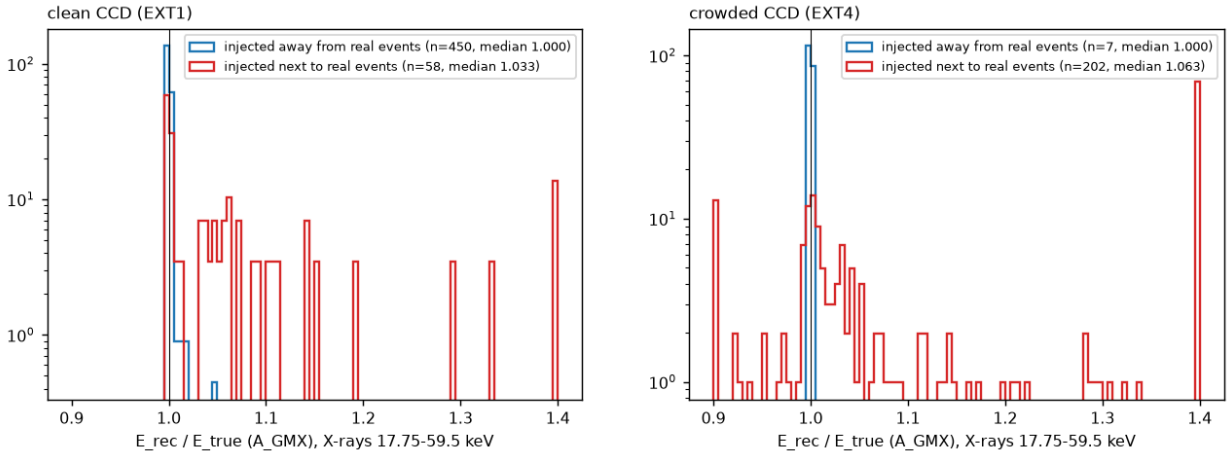


Figure 25: $E_{\mathrm{rec}}/E_{\mathrm{true}}$ of injected X-rays (17.75–59.5 keV) away from and next to real events.

15.2 Where the excess comes from

Two mechanisms, both real charge of other deposits:

1. **A charge sheet under the cluster** (dominant on the crowded CCD). Many small overlapping deposits make a diffuse sheet of 20–200 e^-/pix. Sharing gives the part of the sheet under a cluster's footprint to the cluster; the cluster still looks clean (one cloud, good χ^2, blend ≈ 0). The median charge in a ring 5–8 px around 59.5-keV-like clusters is 24 e^-/pix for clusters at the peak but 116–218 e^-/pix for clusters at 60.5–66 keV (figure 26). WADERs mostly avoids it because it only sums core pixels above $\sim$80 e^-.

2. **A neighbouring deposit merged into the cluster** (smaller effect on real data): such clusters have several clouds and a bad fit of their own model (median χ^2 15 vs 1; figure 27).

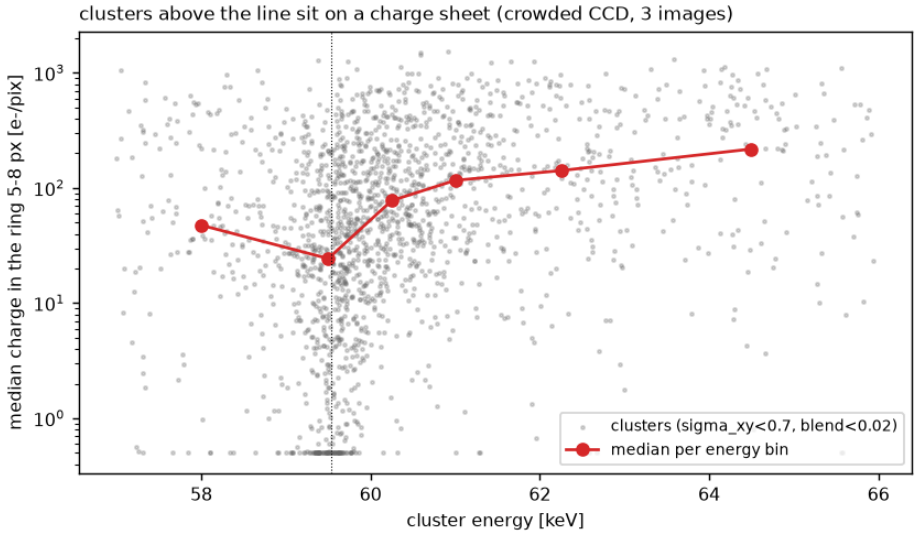


Figure 26: Charge in a ring 5–8 px around each 59.5-keV-like cluster against its energy: clusters above the line sit on a charge sheet.

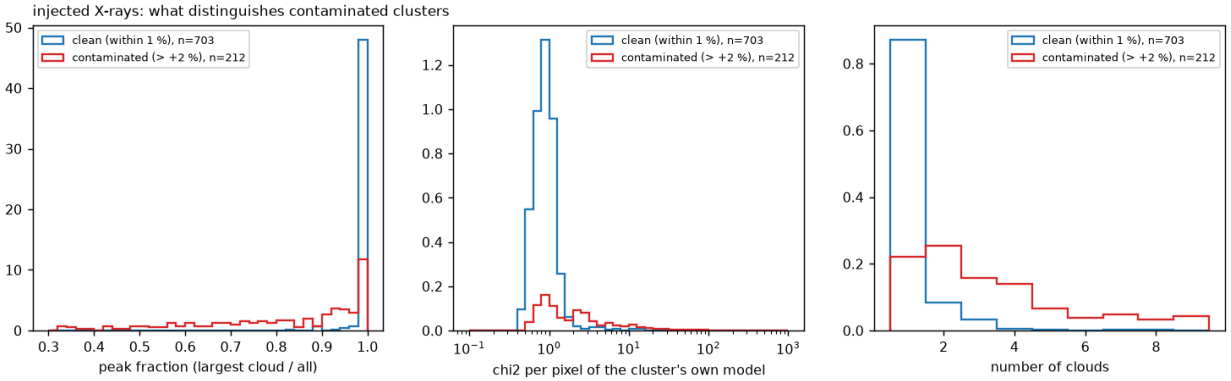


Figure 27: Injected X-rays: peak fraction, χ^2 of the cluster's own model and number of clouds for clean (within 1 %) and contaminated (> +2 %) clusters.

15.3 What does not work: re-measuring the energy

Growth off, pixel clipping at model +3–4σ, a local background from unexplained pixels, a core aperture (2 or 3 σ) with a model correction, a local pedestal from the ring (median or 25th percentile) subtracted over the footprint, and a joint fit of the cluster's shape + a flat level were all tested on the injections and the real lines. They either change nothing or add so much noise (the sheet is not uniform) that the lines broaden: e.g. subtracting the ring median removes the tail (0.60 → 0.22) but destroys the Np L fit (Z 34 → 5, $k = 0.965$).

15.4 What works: flags and a selection

Energies are left unchanged; every cluster now carries three quality flags (`gmx.general_cluster`):

$$\mathrm{ped}_c = P_{25}\{R_p : 5 \le |\mathbf{x}_p - \mathbf{c}| \le 8\ \mathrm{px}\}, \qquad \mathrm{peakfrac}_c = \frac{\max_{k\in c} A_k}{\sum_{k\in c} A_k}, \qquad \chi^2_c = \frac{1}{n_c} \sum_{p:\ \tilde{r}_{pc}>0.5} \frac{(R_p - M_p)^2}{V_p + M_p}, \tag{19}$$

the local pedestal (25th percentile of the residual charge in the ring; robust against neighbours in the ring), the share of the cluster's fitted charge in its largest cloud, and the fit quality of its own model. On injected X-rays, peakfrac > 0.95 keeps 98 % of clean clusters and rejects 67 % of contaminated ones; on the real crowded CCD the pedestal is what matters (figure 28):

selection (crowded CCD, 29 images)	clusters	59.5 tail/peak	k	dev/ndf	Z	Lβ / Lγ / γ26.3 / Ag K
WADERs ($\sigma_{xy} < 0.7$)	62 665	0.10	0.984	1.74	27.1	15.5 / 16.0 / 14.8 / 12.9
A_GMX ($\sigma_{xy} < 0.7$, blend < 0.02)	65 876	0.60	1.0014	1.98	34.3	19.9 / 19.1 / 19.9 / 18.3
+ ped < 10 e^-/pix	43 143	0.22	0.9995	1.60	28.7	16.1 / 16.0 / 17.3 / 15.6
+ ped < 5 e^-/pix	37 507	0.18	0.9994	1.54	26.9	14.7 / 15.3 / 16.2 / 14.6

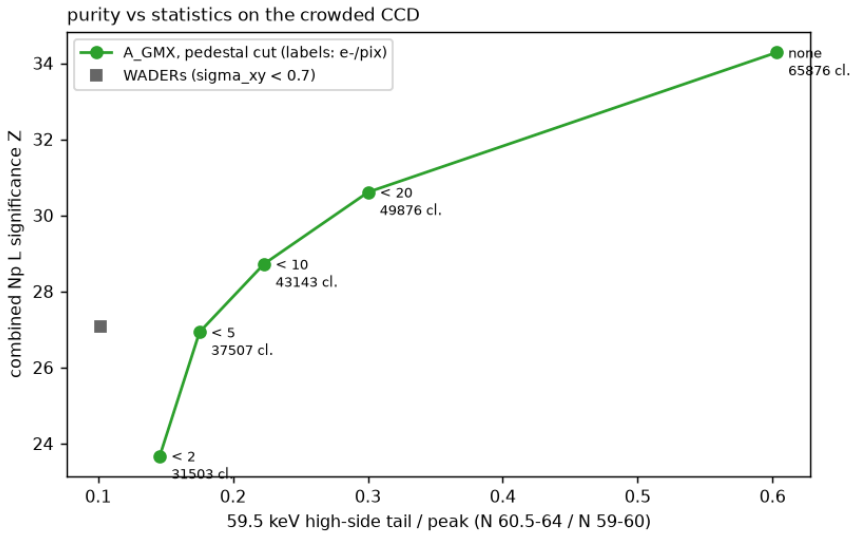


Figure 28: Purity against statistics: the 59.5 keV tail and the Np L significance for pedestal cuts of 20, 10, 5, 2 e^-/pix.

Satellites. Satellite merging absorbed any cluster with $< 1\,\%$ of a neighbour's energy, so a 5.9-keV X-ray next to a > 600-keV deposit was swallowed (crowded CCD, injected: median $E_{\mathrm{rec}}/E_{\mathrm{true}}$ 8.7). Only fragments below 1 keV are merged now (`sat_e_max`); the X-ray stays its own cluster (unit test `test_xray_next_to_big_deposit_not_merged`), with an energy a few % low when it sits on the big deposit's wing.

Recommended selection for spectroscopy. $\sigma_{xy} < 0.7$ (or the front-side cut of the analysis), blend < 0.02, ped $< 10\,e^-$/pix, not on the edge; add peakfrac > 0.95 where merged neighbours matter. The flags are outputs: nothing is removed by `general_cluster` itself.

15.5 Effect on all X-ray lines and on Fe-55

Every Am-241 line fitted alone (Gaussian + linear background) on EXT1, EXT3 and EXT4 of the 29-image run, old and new code, with and without the flags (`am241_lines_flags.py`), and the front-side Fe-55 Kα of every data set of the wide test (`wide_agmx_flags.py`, `summ_wide_flags.py`):

CCD	selection	clusters	Np La	Np Lb	Np Lg	γ 26.3	γ 59.5	59.5 tail
EXT1	old A-GMX (sxy<0.7, blend<0.02)	13632	–	–	–	–	0.9867 (206)	0.23
EXT1	new A-GMX (sxy<0.7, blend<0.02)	13632	–	–	–	–	0.9867 (206)	0.23
EXT1	new + ped<10	13632	–	–	–	–	0.9867 (206)	0.23
EXT1	new + peakfrac>0.95	13520	–	–	–	0.9722 (256)	0.9867 (208)	0.23
EXT1	new + ped<10 + peakfrac>0.95	13520	–	–	–	0.9722 (256)	0.9867 (208)	0.23
EXT3	old A-GMX (sxy<0.7, blend<0.02)	75580	–	0.9943 (200)	–	–	0.9989 (246)	0.11
EXT3	new A-GMX (sxy<0.7, blend<0.02)	82431	–	–	–	–	0.9989 (246)	0.11
EXT3	new + ped<10	80808	–	–	–	–	0.9989 (243)	0.11
EXT3	new + peakfrac>0.95	66321	–	0.9956 (200)	–	–	0.9993 (246)	0.14
EXT3	new + ped<10 + peakfrac>0.95	64886	–	0.9957 (200)	–	–	0.9993 (242)	0.13
EXT4	old A-GMX (sxy<0.7, blend<0.02)	65876	–	–	–	1.0038 (273)	1.0017 (353)	0.60
EXT4	new A-GMX (sxy<0.7, blend<0.02)	81660	–	0.9996 (266)	–	1.0040 (279)	1.0017 (353)	0.62
EXT4	new + ped<10	48613	–	0.9974 (200)	–	1.0007 (242)	1.0006 (317)	0.22
EXT4	new + peakfrac>0.95	70730	–	1.0004 (270)	1.0049 (245)	1.0045 (286)	1.0025 (350)	0.68
EXT4	new + ped<10 + peakfrac>0.95	41428	–	0.9978 (203)	–	1.0009 (248)	1.0014 (321)	0.23

Every Am-241 line fitted alone: peak / true energy (width in eV); Np Lβ and Lγ are groups of lines, so their centres are not expected at exactly 17.75 / 20.78 keV (the constrained fit's k is the check there).

data set	n	Ka old (all)	Ka new (all)	front	front+ped	front+pf	front+b
2025-08-27-DM07_DM09_DM10_Low_	102	5.8145	5.8145	5.7685 (63)	5.7685 (63)	5.8025 (63)	5.8025
2025-08-30-DM11_DM13_DM14_Low_	102	5.7869	5.7868	5.7616 (69)	5.7616 (69)	5.7736 (66)	5.7736
2025-09-03-DM15_DM16_DM17_Low_	102	5.7766	5.7760	5.7465 (67)	5.7465 (67)	5.7597 (66)	5.7597
2025-09-06-DM18_DM19_DM20_Low_	102	5.7605	5.7603	5.7284 (69)	5.7285 (69)	5.7370 (68)	5.7370
2025-09-07-DM18_DM19_DM20_Low_	18	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-09-09-DM21_DM22_DM23_Low_	18	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-09-13-DM24_DM25_DM26_Low_	90	5.9023	5.9020	5.8826 (66)	5.8826 (66)	5.8755 (66)	5.8755
2025-09-16-DM08_DM11_DM12_Low_	18	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-09-19-DM07_DM27_DM28_Low_	18	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-10-31-DM04_DM05_DM06_Low_	27	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-11-04-DM10_DM24_DM25_Low_	36	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-11-08-PD06_PD07_PD08_Low_	36	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n
2025-11-13-DM01_DM02_DM03_Low_	36	nan	nan	nan (nan)	nan (nan)	nan (nan)	nan (n

Fe-55 K-alpha per data set: mean fitted peak [keV] (median width eV); old vs new A_GMX without cuts, new front side with the flag cuts

selection	image-CCDs	mean K-alpha	spread	width
old A_GMX, all clusters	401	5.8057	151	79
new A_GMX, all clusters	401	5.8055	151	79
new, front	380	5.7757	155	67
new, front, ped<10	380	5.7757	155	67
new, front, peakfrac>0.95	345	5.7900	156	66
new, front, ped<10, peakfrac>0.95	345	5.7900	156	66

Fe-55 K-alpha, all Fe-55 image-CCDs: number fitted, mean peak [keV], spread over image-CCDs [eV], median width [eV]

data sets	image-CCDs	Fe-55	muon	worm	pair 4 px
Fe-55	705	0.951 / 0.955	0.672 / 0.672	0.829 / 0.830	0.912 / 0.917
Am-241	115	0.819 / 0.828	0.424 / 0.422	0.585 / 0.587	0.622 / 0.637
skip-noise	58	0.799 / 0.799	0.675 / 0.680	0.780 / 0.780	0.899 / 0.904

Injection test (same objects): success old / new A_GMX

16 Does A_GMX supersede the other methods?

Yes, as the general clusterer. Against each alternative:

method	where it is still better than A_GMX	verdict
MHC	Fe-55 data sets: Fe-55 success 0.964 vs 0.951 (energies equal); hot columns 0.78 vs 0.74; 4-px pairs on Am-241 sets 0.83 vs 0.62 (A_GMX hands the noisy PD06 images to hysteresis)	**superseded.** A_GMX is better on crowded images (Lβ 19.9 vs 8.0σ), noisy images (0.86 vs 0.61), high dark charge, worms, alphas, 3-px pairs (0.75 vs 0.29), diffuse 5-px pairs (0.93 vs 0.77) and energy resolution (0.19 vs 0.21 %), and is faster (2× on typical images, 3× on the Am-241 frames, ∼300× on the noisy PD06 frames). MHC's model-selection idea lives on in GMX's island test. Keep the code as a cross-check.
hybrid v1 / v2	nothing significant	**superseded** (they merge diffuse pairs and fail on crowded CCDs).
adaptive (MHC engine)	nothing significant	**superseded**: same switch, weaker engine (ties WADERs on the crowded CCD instead of beating it).
GMX alone	identical except on "unmodelled" images, where it fails (Fe-55 0.49)	**superseded** by A_GMX (GMX + the switch).
MFD+G	isolated X-rays only: Fe-55 success 0.972 vs 0.951, 5× faster	**keep for Fe-55 gain calibration** (energies identical; only point-like X-rays matter). Not for tracks (muon 0.18).
B auto	as fast as WADERs (100× faster than A_GMX); better than WADERs everywhere except the noisy PD06 images	**keep** as the cheap pipeline option (`--cluster-threshold-e auto`).
WADERs	noisy "unmodelled" images (where A_GMX *is* WADERs); speed; the established pipeline	**keep as the reference**; it is what A_GMX falls back to.

In one line: use A_GMX for analysis (it is never much worse than the best method in any regime and is the best in most); use MFD+G when you only need the Fe-55 peak fast; keep WADERs / B for the standard pipeline until A_GMX is integrated and checked by eye on real tracks.

17 Limits and open items

- **Very sharp clusters:** the pixel-integration approximation ($\Sigma + \frac{1}{12}$) over-predicts the peak for $\sigma \lesssim 0.2$ px; sharing then moves a few % of a sharp X-ray's charge to a diffuse neighbour (worked example). An exact pixel integral (error functions, already in `agent_methods.gauss_pix`) would fix it at some cost in speed.
- **Unmodelled images** fall back to WADERs: close pairs are then not resolved (0.15). A background model for those images would let GMX run there.

- **EM depends on the seeds** (scale-space seeding is what makes close pairs work); no global optimum is guaranteed.
- **Island test** tolerates 5 % non-Gaussianity; only checked on injected objects and Fe-55. **Real Am-241 tracks have not been inspected by eye.**
- **Labels:** muon/worm swap ∼30 %; classes only validated on injections.
- **Crowded fit quality:** dev/ndf 1.98 with structure at 25–28 keV; the σ_{xy} cut is not re-optimised for A_GMX's width.
- **Speed:** ∼40 s on a crowded 1.75-Mpix CCD (WADERs: < 1 s).
- Not integrated in `run_WADERs` (would be an optional flag; needs approval).

18 Parameters, code, how to run

parameter	default	where	meaning
`excess_ratio`	3	adaptive	unmodelled if $s \geq 3\sigma_{\text{read}}$ (and ≥ 0.
`occ_thr`, `box`	0.3, 31 px	adaptive	"crowded" label (information)
`scales`	0.5, 1, 1.5, 2.5 px	GMX	matched-filter widths
`nsig`	5	GMX	seed significance (+ Poisson check
`n_iter`	30 (+15 residual, +5 pruning)	GMX	EM iterations
`nsig_res`	5	GMX	residual-pass seeds
`sig_min`	4	GMX	prune clouds with S/N below
`r_merge`	0.5	GMX	ridge ratio for merging
`w_ratio`	1.6	GMX	width ratio (sharp-on-diffuse prote
`link_alpha`, `island_max`, `bic_margin`	0.01, 4000 px, 30	GMX	island test
`f_sat`, `sat_e_max`	0.01, 1 keV	GMX	satellite absorption (only fragment
`ped_ring`, `ped_q`	(5, 8) px, 25	GMX	local pedestal flag (ring, percentile
`grow`	3 px	GMX	CTI-trail growth
`p_count`, `count_E_max`	10^{-5}, 100 e^-	GMX	few-electron finder, counting energ

```
import sys; sys.path.insert(0, "comm_scripts/clustering_tests")
import adaptive as AD
clusters = AD.adaptive_cluster(Q_electrons, valid_mask, engine="gmx")
# each cluster: E [keV], y, x, sxy [px], cls, blend, sig, n_comp, edge, ped [e-/pix], peakfrac, chi2,
# regime ('sparse' / 'crowded' / 'unmodelled'), occ
# spectroscopy selection (section on contamination):
good = [c for c in clusters if c.get("blend", 0) < 0.02 and not c.get("edge")
       and not c.get("ped", 0) >= 10 # not sitting on a charge sheet (crowded fields)
       and c.get("peakfrac", 1) > 0.95] # not merged with a neighbouring deposit
```

Test scripts (all in `clustering_tests/`, README there): `wide_test.py` + `wide_run.py` `adaptive+GMX` (all images), `adaptive_validate.py` (tracks, few-electron, Fe-55, crowded), `make_agmx_figures` `make_agmx_perf.py`. Unit tests: `python -m unittest tests.test_clustering_methods tests.test_agmx_`

18.1 Running A_GMX on WADERs data: cluster files in the WADERs format

A_GMX is part of the `run_WADERs` chain as an option, and can be run on a finished run:

```
run_WADERs images.txt --gain-correction --clusterer agmx --composite # A_GMX files, composite on them
run_WADERs images.txt --gain-correction --clusterer both --composite # also composite_agmx/ next to co
run_AGMX WADERs_analysis_<run> [-j 4] [--per-image] [--no-gaincorr] # a run that is already finished
```

It runs on the calibrated image panaSKImg saves for every CCD (`<image>_EXT<n>_waders.fits.bz2`). With the gain correction, the image is first converted ADU $\rightarrow e^-$ with the same $g(n)$ curves and anchor as `root_gaincorr/`; a CCD without a reliable curve keeps its fitted gain, as there. The output is `root_agmx/` and `root_agmx_gaincorr/`: one `panaSKImg_clustersRec_<image>_ext<n>.root` per CCD, written by `src/agmx_root.py`. Every A_GMX cluster (its pixels and the charge it takes from each, which sums to E) becomes a pysimdamicm `Cluster` object, the ClusterFinder's own class, so all branches (positions, `wSTD`, `fSTD`, PCA, track end points, seed, `closest_cls_*`, `Energy_PS/AVG`, …) are computed by panaSKImg's code. `info` and `process_config` are copied from the CCD's panaSKImg file, and the extra branches `agmx_cls`, `agmx_blend`, `agmx_ped`, `agmx_peakfrac`, `agmx_chi2`, `agmx_sig`, `agmx_regime` and a `TNamed` `agmx_config` are added. MakeComposite and every other reader of WADERs files read them unchanged.

Check. `clustering_tests/agmx_root_check.py` writes the clusters of real panaSKImg files (EXT1, EXT3 and EXT4 of an Am-241 image: 1652, 7686 and 5452 clusters) through the writer and compares every branch. They are identical except in numerically degenerate cases, where panaSKImg's own result depends on the last bits of its inputs:

- Gaussian fits (`fSTD`) of profiles 1–3 bins wide;
- the sign of a PCA eigenvector (θ versus $\theta \pm \pi$);
- PCA of 1–2 pixel clusters;
- `closest_cls_*` when the nearest cluster lies in the same row or column, because panaSKImg excludes "itself" with `PosX != ... and PosY != ...`.

With the default `--clusterer waders` the pipeline output is identical to before (`utils/compare_runs.py`). A flag value of -999 in the files means "not computed" (e.g. `agmx_chi2` of a cluster the mixture did not fit). Every cluster carries the flags: clusters from the few-electron finder and the hysteresis branch get `edge` and `ped` like the others, `blend` $= 0$ and `peakfrac` $= 1$ (no charge is shared).

Fixes of 25 October (audit). (1) The peak search ignored masked pixels only after the local-maximum test, so the matched-filter significance just outside the mask (variance 0 there) could hide a peak on the first or last valid row or column: such an X-ray came out as a flag-less few-electron cluster. The local maximum is now taken over valid pixels only, and the X-ray is a normal cluster with `edge` set. (2) The Poisson check of residual few-electron seeds removed the dark charge twice; with it removed once, a simulation at $\lambda = 0.5\,e^-$/pix finds 20 instead of 14 of 60 injected $15\,e^-$ events, with no clusters on pure-noise images either way. Both change only events touching the mask and high-dark-current CCDs; the performance numbers of this document were made before them.

A All performance tables

Generated by `clustering_tests/make_agmx_perf.py` from the saved results (same tests as section 14; all methods; "ok" = one cluster with energy within 5 %, pairs: both resolved).

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.856	0.52 %	0.59	0.74	0.11	0.01
B auto	0.905	0.59 %	0.62	0.78	0.13	0.01
MFD+G	0.933	0.21 %	0.18	0.37	0.94	0.14
hybrid v1	0.873	0.20 %	0.58	0.66	0.84	0.02
hybrid v2	0.930	0.21 %	0.63	0.68	0.85	0.05
MHC	0.933	0.21 %	0.64	0.77	0.86	1.61
adaptive (MHC engine)	0.931	0.21 %	0.64	0.77	0.83	1.26
GMX	0.913	0.19 %	0.63	0.78	0.89	0.65
A_GMX	0.925	0.19 %	0.64	0.79	0.87	0.81

Wide test, all (878 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.880	0.47 %	0.62	0.78	0.11	0.01
B auto	0.927	0.53 %	0.65	0.81	0.13	0.01
MFD+G	0.972	0.17 %	0.20	0.39	0.95	0.13
hybrid v1	0.926	0.16 %	0.63	0.71	0.87	0.02
hybrid v2	0.972	0.17 %	0.68	0.73	0.88	0.04
MHC	0.964	0.17 %	0.68	0.81	0.87	0.94
adaptive (MHC engine)	0.961	0.17 %	0.67	0.81	0.87	0.83
GMX	0.951	0.16 %	0.67	0.83	0.91	0.57
A_GMX	0.951	0.16 %	0.67	0.83	0.91	0.64

Wide test, Fe-55 data sets (705 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.830	0.77 %	0.42	0.60	0.12	0.11
B auto	0.822	0.89 %	0.42	0.60	0.14	0.24
MFD+G	0.738	1.21 %	0.11	0.27	0.88	3.79
hybrid v1	0.605	0.78 %	0.29	0.37	0.69	8.10
hybrid v2	0.728	1.18 %	0.37	0.46	0.72	16.89
MHC	0.793	1.11 %	0.41	0.53	0.83	76.69
adaptive (MHC engine)	0.794	0.95 %	0.41	0.56	0.61	38.31
GMX	0.730	0.75 %	0.38	0.50	0.74	39.36
A_GMX	0.819	0.72 %	0.42	0.58	0.62	22.73

Wide test, Am-241 data sets (115 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.567	1.59 %	0.50	0.53	0.10	0.01
B auto	0.784	1.58 %	0.64	0.76	0.18	0.03
MFD+G	0.824	1.18 %	0.14	0.29	0.97	0.32
hybrid v1	0.738	1.05 %	0.52	0.60	0.79	0.07
hybrid v2	0.804	1.17 %	0.55	0.58	0.81	0.21
MHC	0.817	1.17 %	0.68	0.72	0.86	10.70
adaptive (MHC engine)	0.812	1.17 %	0.68	0.71	0.85	7.77
GMX	0.799	1.19 %	0.68	0.78	0.90	1.00
A_GMX	0.799	1.19 %	0.68	0.78	0.90	1.08

Wide test, skip-noise study (58 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.867	1.66 %	0.48	0.66	0.08	2.40
B auto	0.802	2.08 %	0.39	0.65	0.15	2.46
MFD+G	0.281	9.15 %	0.08	0.09	0.73	69.85
hybrid v1	0.275	8.24 %	0.13	0.10	0.67	98.38
hybrid v2	0.278	8.99 %	0.15	0.10	0.68	156.01
MHC	0.608	5.62 %	0.45	0.45	0.81	396.59
adaptive (MHC engine)	0.864	2.02 %	0.50	0.70	0.15	1.29
GMX	0.488	4.45 %	0.31	0.36	0.65	226.14
A_GMX	0.864	2.02 %	0.50	0.70	0.15	1.25

Wide test, noisy (bg spread $\geq$ 1 e-) (27 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.550	2.40 %	0.33	0.45	0.07	0.08
B auto	0.806	2.10 %	0.48	0.66	0.17	0.08
MFD+G	0.772	1.07 %	0.15	0.29	0.91	0.70
hybrid v1	0.724	1.02 %	0.39	0.51	0.82	0.11
hybrid v2	0.765	1.06 %	0.44	0.52	0.83	0.32
MHC	0.829	1.00 %	0.54	0.64	0.88	16.18
adaptive (MHC engine)	0.868	0.94 %	0.55	0.68	0.75	7.31
GMX	0.804	0.88 %	0.52	0.67	0.88	1.87
A_GMX	0.871	0.90 %	0.55	0.74	0.79	1.62

Wide test, dark $\geq$ 1 e-/pix (152 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.590	1.64 %	0.27	0.43	0.08	0.08
B auto	0.781	1.60 %	0.39	0.58	0.15	0.18
MFD+G	0.789	1.05 %	0.12	0.29	0.90	1.55
hybrid v1	0.684	0.84 %	0.32	0.44	0.74	0.36
hybrid v2	0.781	1.02 %	0.39	0.50	0.77	2.43
MHC	0.822	0.98 %	0.44	0.58	0.85	25.13
adaptive (MHC engine)	0.819	0.92 %	0.44	0.59	0.70	11.35
GMX	0.784	0.78 %	0.42	0.60	0.82	6.07
A_GMX	0.839	0.77 %	0.45	0.65	0.75	2.26

Wide test, occupancy ≥ 30 % (194 image-CCDs)

method	sigma 0.3	0.8	1.4	1.8	pair 5 px resolved	real Ka / 5.899	Ka spread
WADERs	0.9974 (0.990)	0.9843 (0.943)	0.9521 (0.837)	0.9214 (0.737)	0.00	0.9928	1.32 %
MFD+G	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9999 (0.995)	0.78	0.9968	0.29 %
MHC	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9998 (0.995)	0.77	0.9969	0.26 %
GMX	1.0000 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	0.93	0.9968	0.23 %
A_GMX	1.0000 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	0.93	0.9968	0.23 %

Fe-55 injection (DM09 / DM10, EXT1 / EXT4, images 0-1): median E_rec/E_true (worst CCD) per width, pairs, real Ka

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93
A_GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93

Injected objects in real Am-241 images, EXT1 (3 images x 3 seeds, success fraction)

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.17	0.81	0.84	0.98	0.97	0.28	0.32	0.17	0.00	0.21	0.90
MFD+G	0.02	0.24	0.92	1.00	0.98	0.10	0.02	0.34	0.22	0.98	1.00
MHC	0.20	0.79	0.92	1.00	0.98	0.10	0.28	0.28	0.22	0.92	1.00
GMX	0.20	0.81	0.89	0.97	0.97	0.28	0.25	0.40	0.78	0.98	1.00
A_GMX	0.20	0.81	0.89	0.97	0.97	0.28	0.25	0.40	0.78	0.98	1.00

Injected objects in real Am-241 images, EXT3 (3 images x 3 seeds, success fraction)

dark e-/pix	method	2 e-	3 e-	4 e-	5 e-	7 e-	fakes/Mpix
0.001	WADERs	0.00	0.00	0.00	0.11	0.29	0.0
0.001	MHC	0.01	0.81	0.97	0.99	1.00	0.0
0.001	GMX	0.01	0.81	0.98	1.00	1.00	0.0
0.001	A_GMX	0.01	0.81	0.98	1.00	1.00	0.0
0.001	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	7.3
0.01	WADERs	0.00	0.00	0.00	0.11	0.30	0.0
0.01	MHC	0.00	0.01	0.38	0.93	1.00	0.0
0.01	GMX	0.00	0.01	0.38	0.93	1.00	0.0
0.01	A_GMX	0.00	0.01	0.38	0.93	1.00	0.0
0.01	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	462.3
0.05	WADERs	0.00	0.00	0.00	0.11	0.35	0.0
0.05	MHC	0.00	0.00	0.00	0.04	0.56	0.0
0.05	GMX	0.00	0.00	0.00	0.04	0.56	0.0
0.05	A_GMX	0.00	0.00	0.00	0.04	0.56	0.0
0.05	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	7794.0

Few-electron events on simulated blanks (efficiency by event charge; fakes = clusters of 2-5 e- per Mpix with nothing injected)

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